

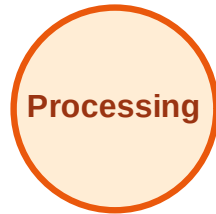
BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing



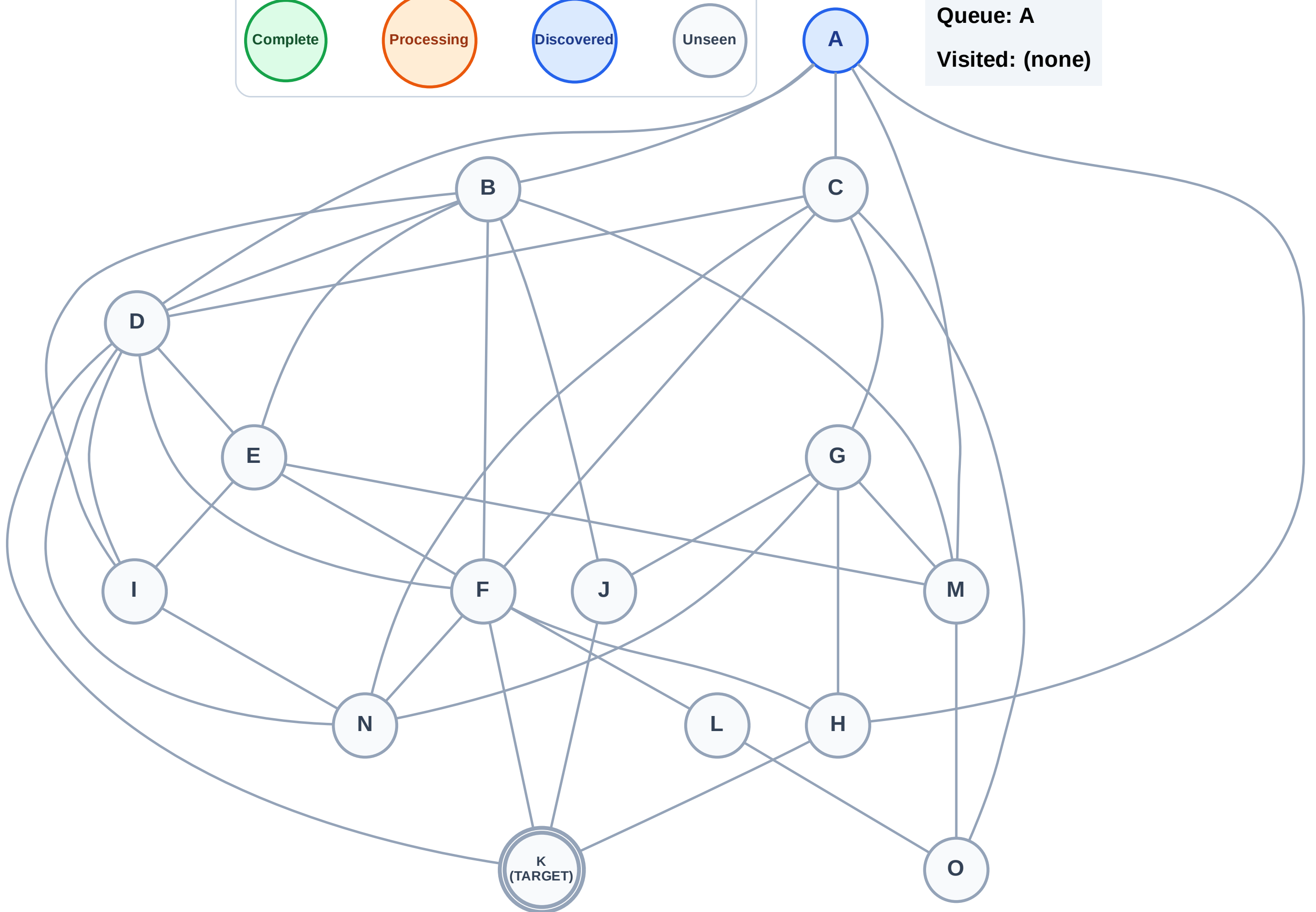
Discovered



Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

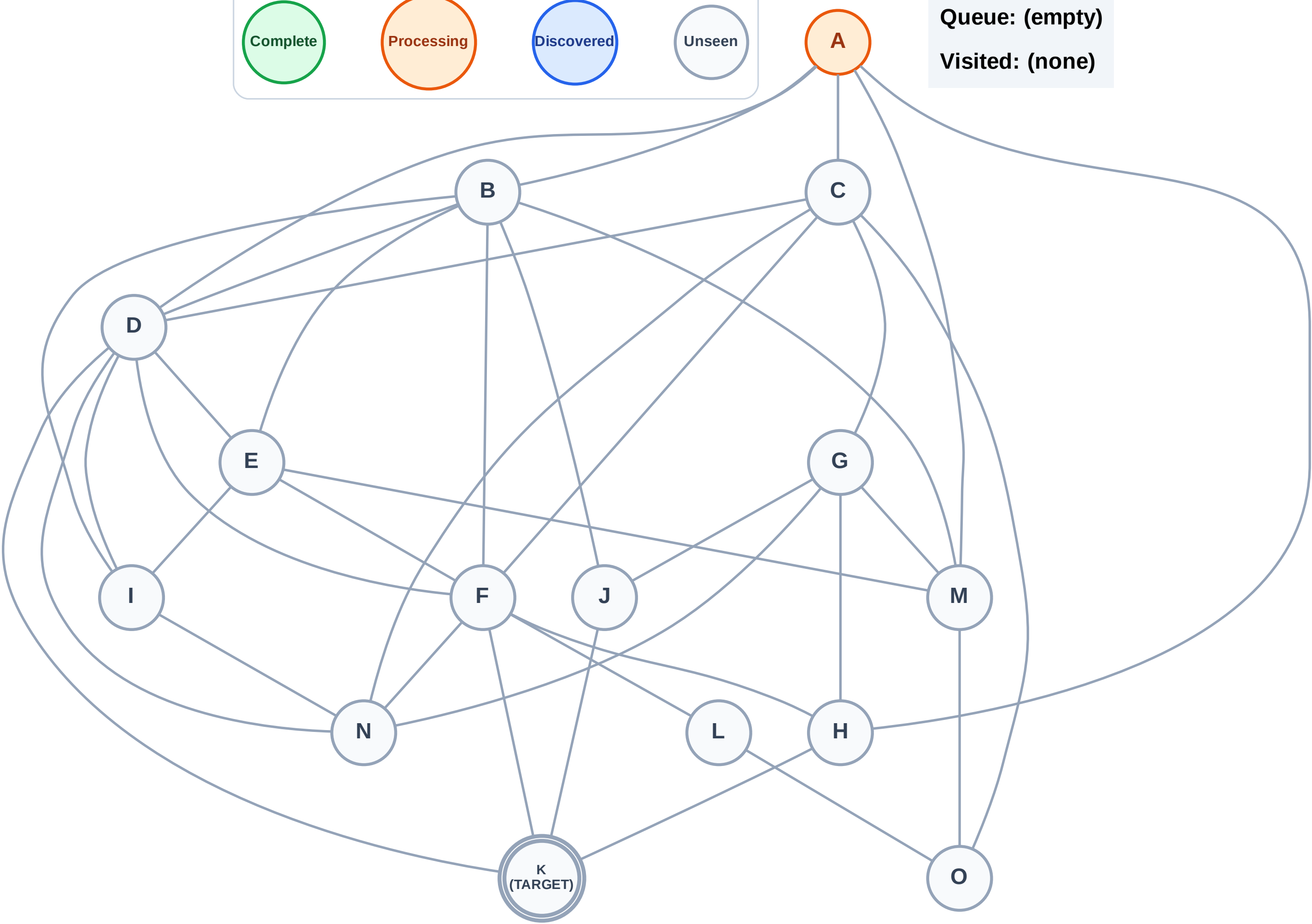
Complete

Processing

Discovered

Unseen

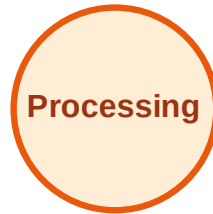
Queue: (empty)
Visited: (none)



BFS Traversal

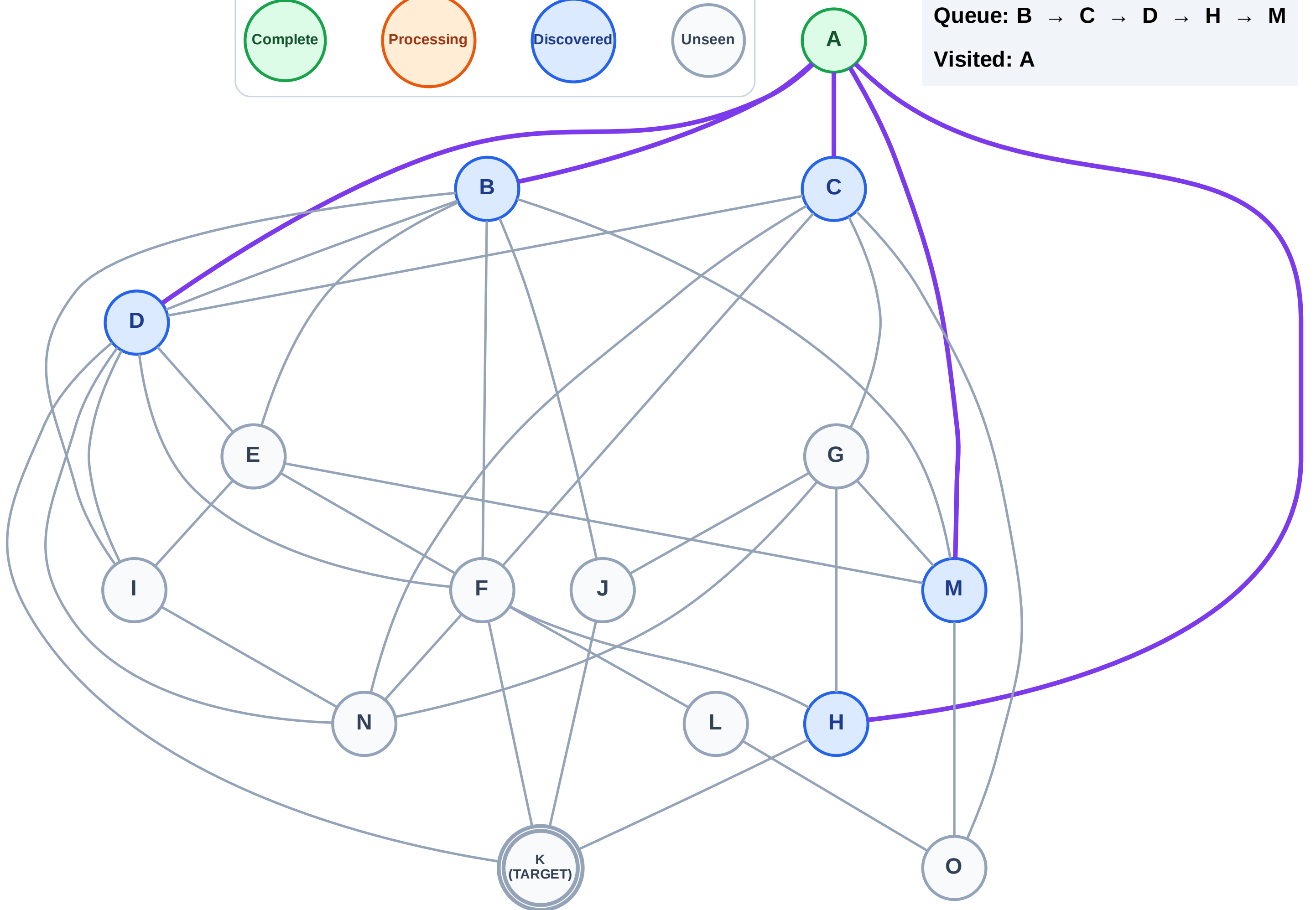
Step 3: Discover B, C, D, H, M from A

Legend



Queue: B → C → D → H → M

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

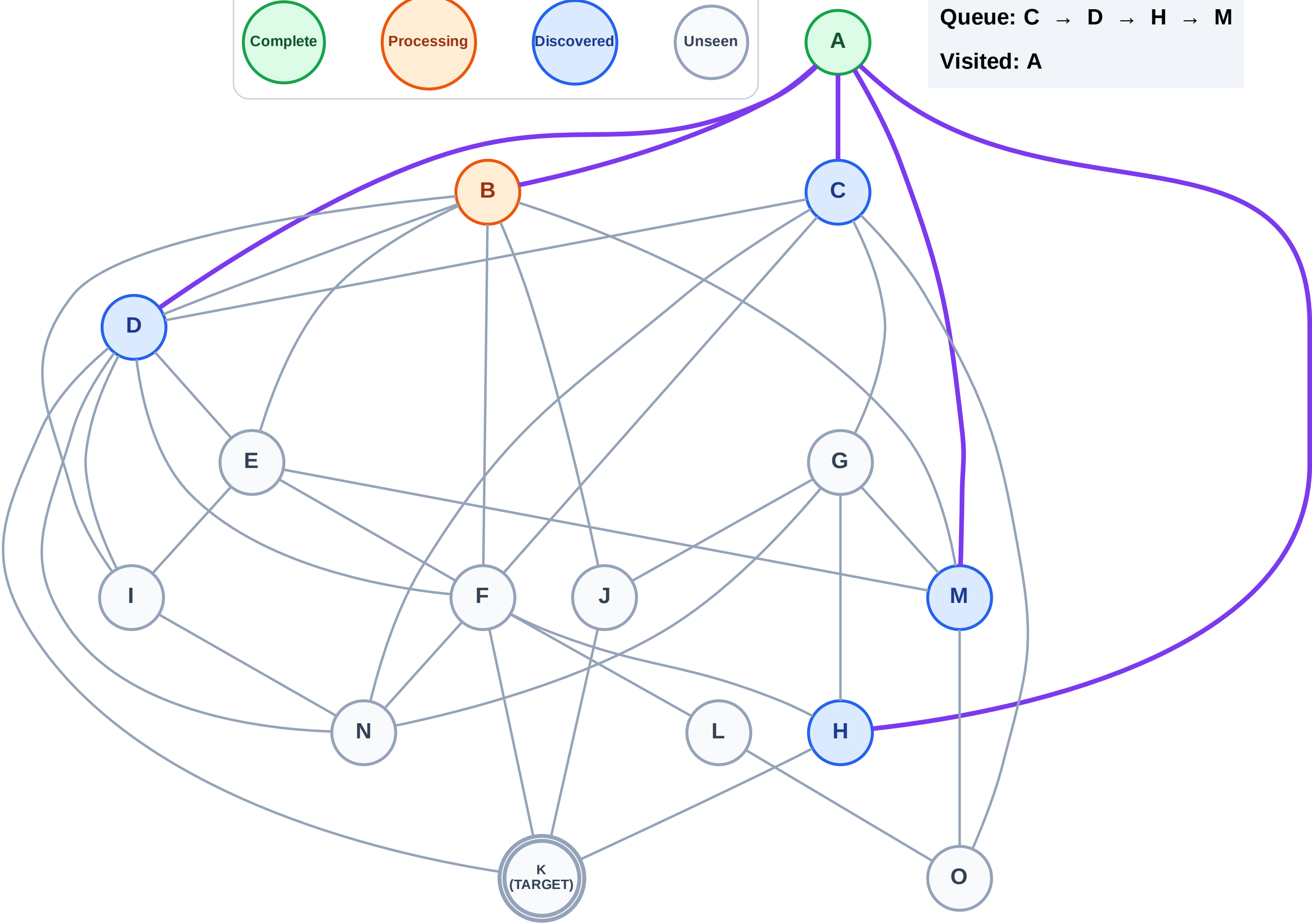
Processing

Discovered

Unseen

Queue: C → D → H → M

Visited: A



BFS Traversal

Step 5: Discover E, F, I, J from B

Legend

Complete

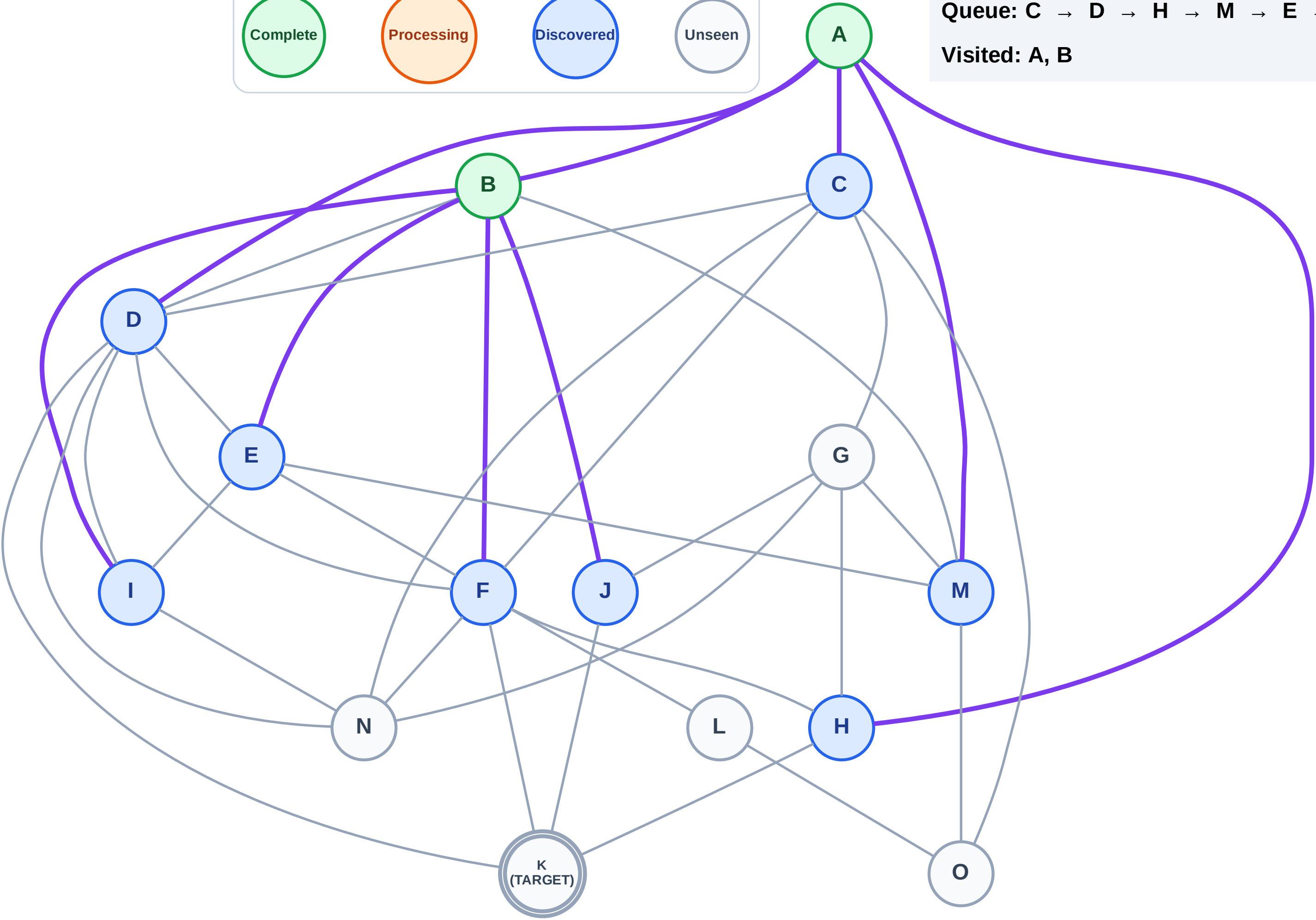
Processing

Discovered

Unseen

Queue: C → D → H → M → E → F → I → J

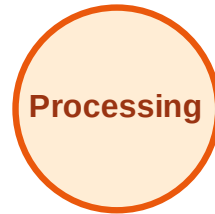
Visited: A, B



BFS Traversal

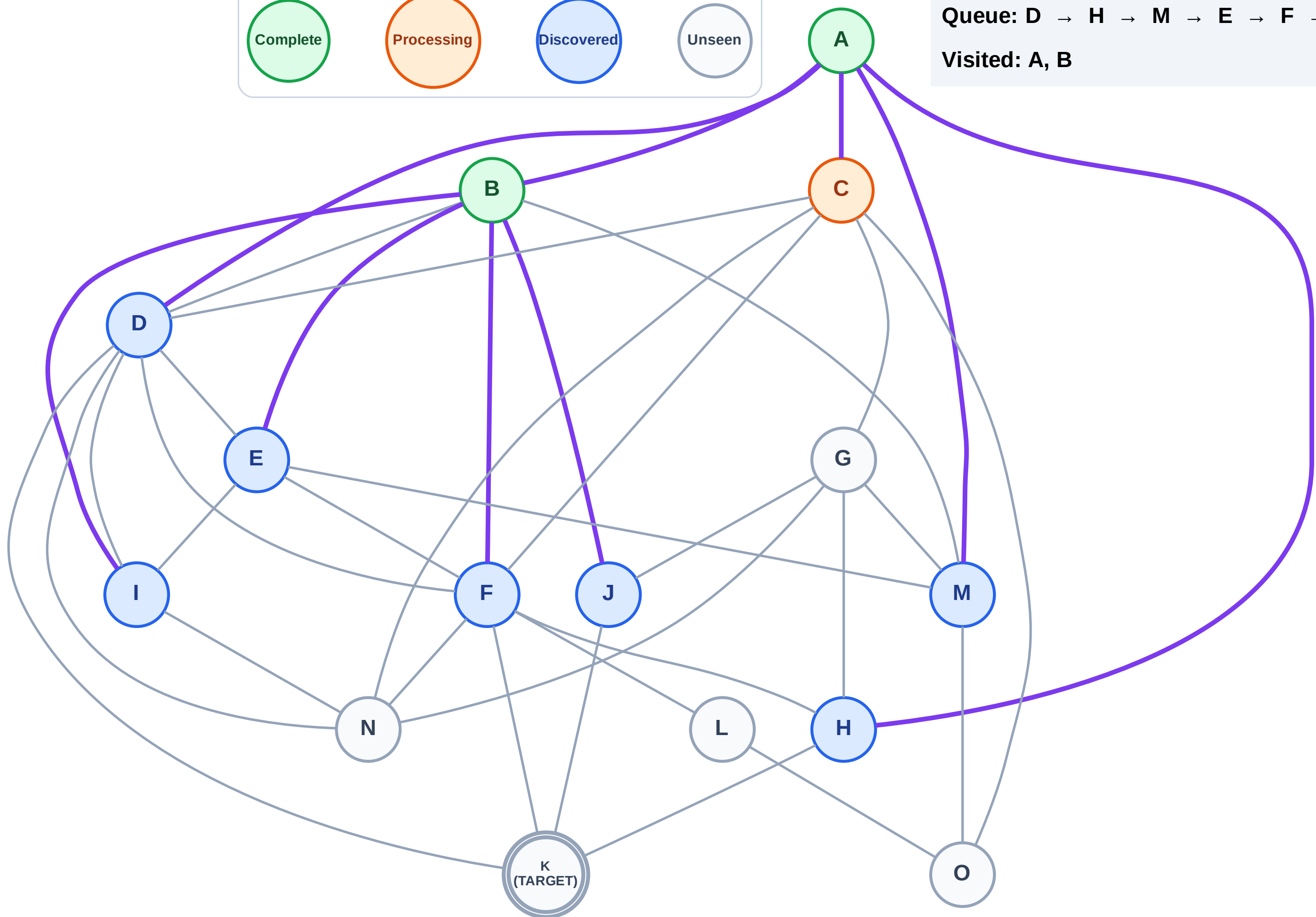
Step 6: Process node C

Legend



Queue: D → H → M → E → F → I → J

Visited: A, B



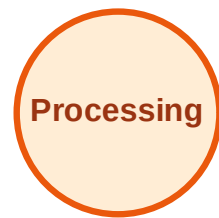
BFS Traversal

Step 7: Discover G, N, O from C

Legend



Complete



Processing



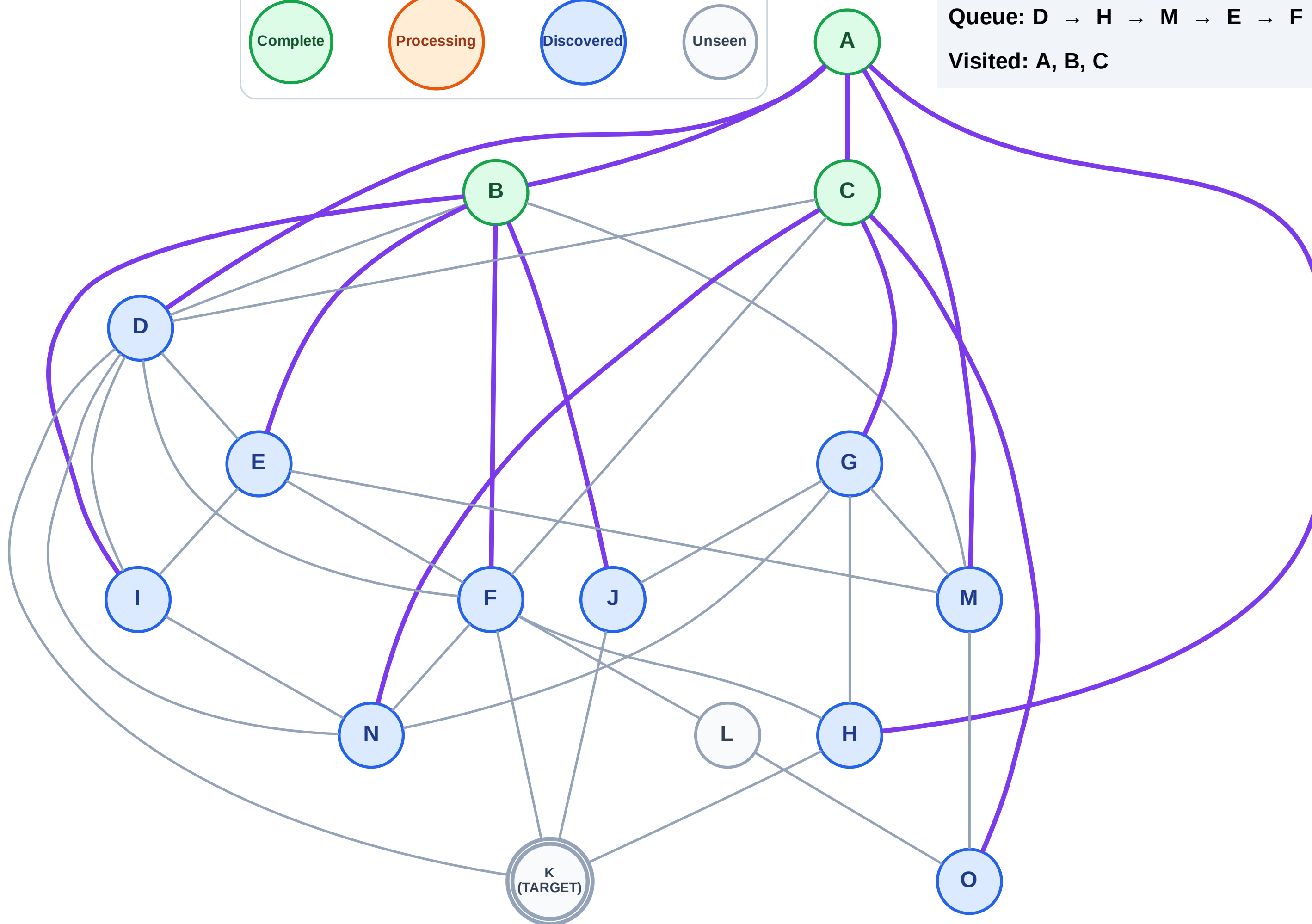
Discovered



Unseen

Queue: D → H → M → E → F → I → J → G → N → O

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

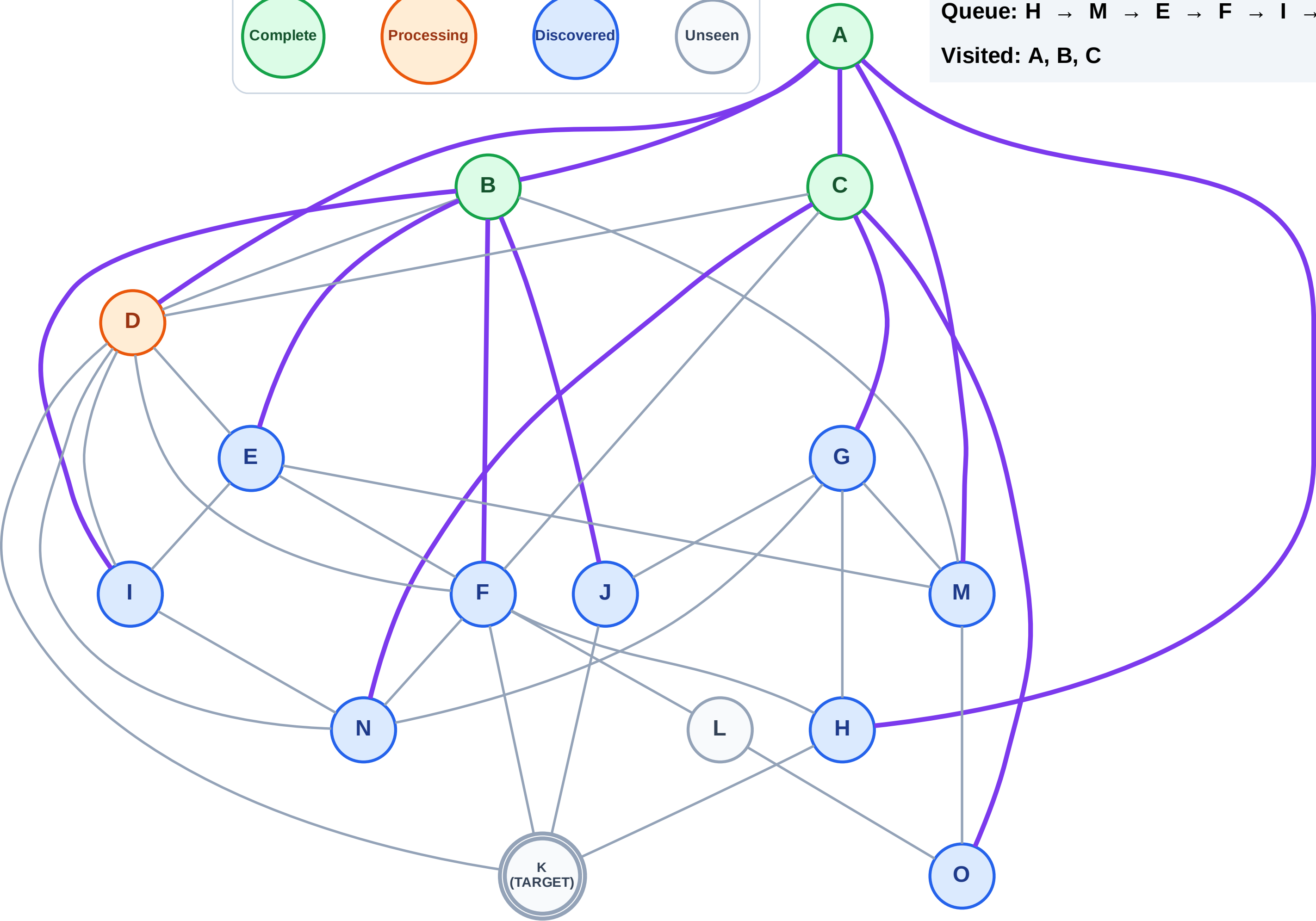
Processing

Discovered

Unseen

Queue: H → M → E → F → I → J → G → N → O

Visited: A, B, C



BFS Traversal

Step 9: Discover K from D

Legend



Complete



Processing



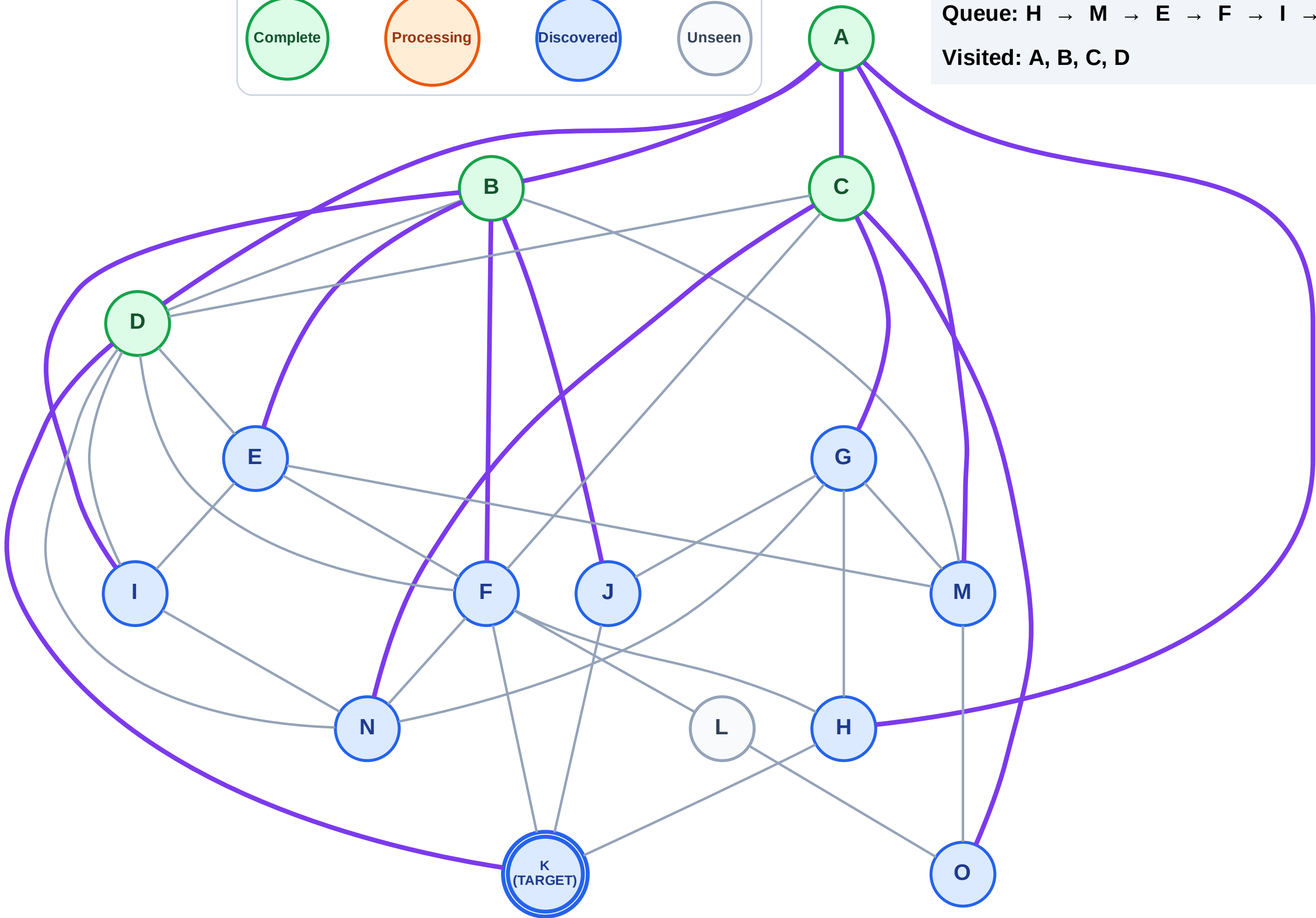
Discovered



Unseen

Queue: H → M → E → F → I → J → G → N → O → K

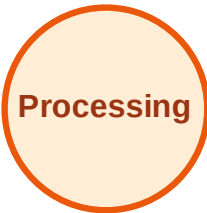
Visited: A, B, C, D



BFS Traversal

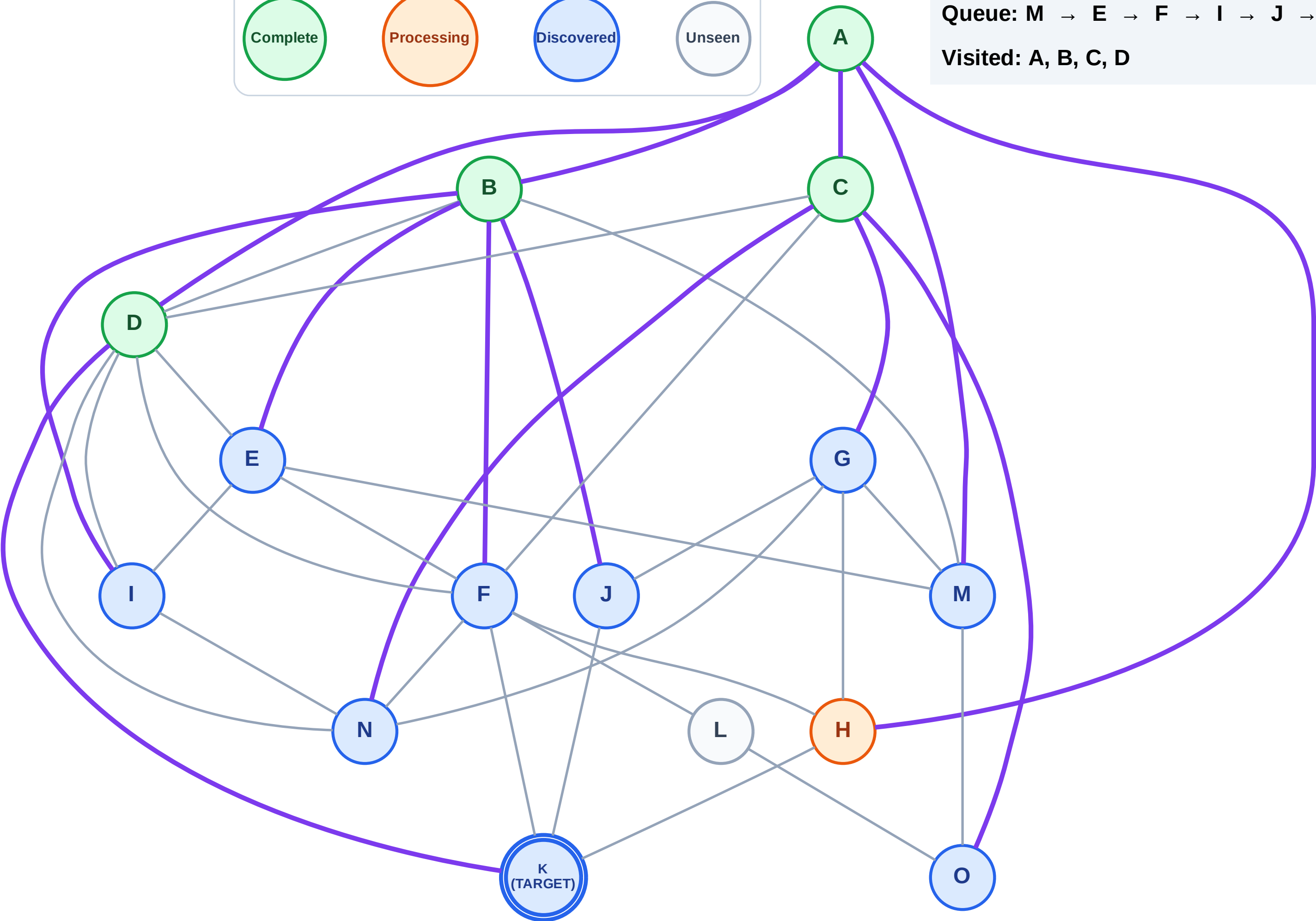
Step 10: Process node H

Legend



Queue: M → E → F → I → J → G → N → O → K

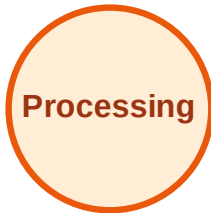
Visited: A, B, C, D



BFS Traversal

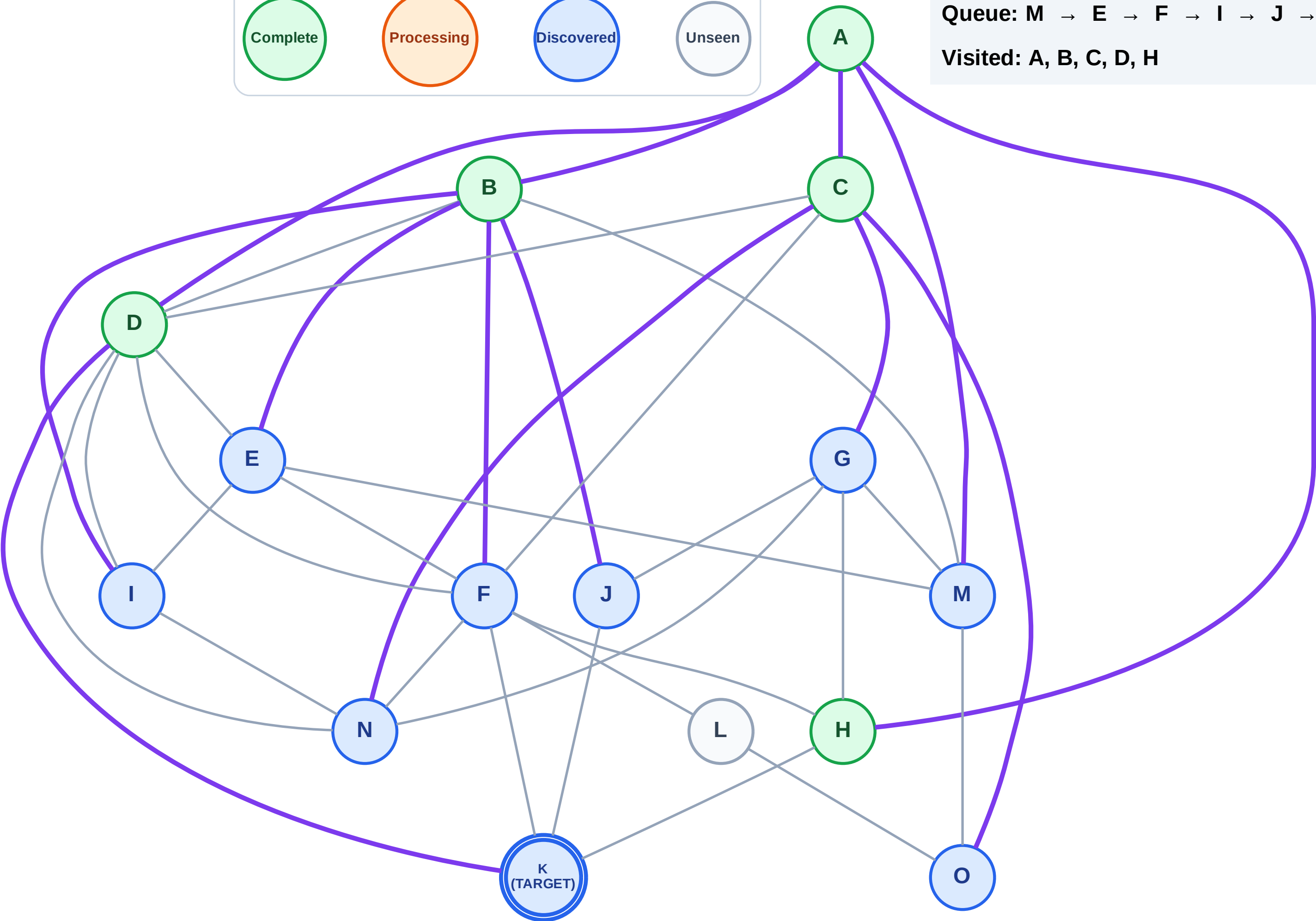
Step 11: No new neighbors from H

Legend



Queue: M → E → F → I → J → G → N → O → K

Visited: A, B, C, D, H



BFS Traversal

Step 12: Process node M

Legend

Complete

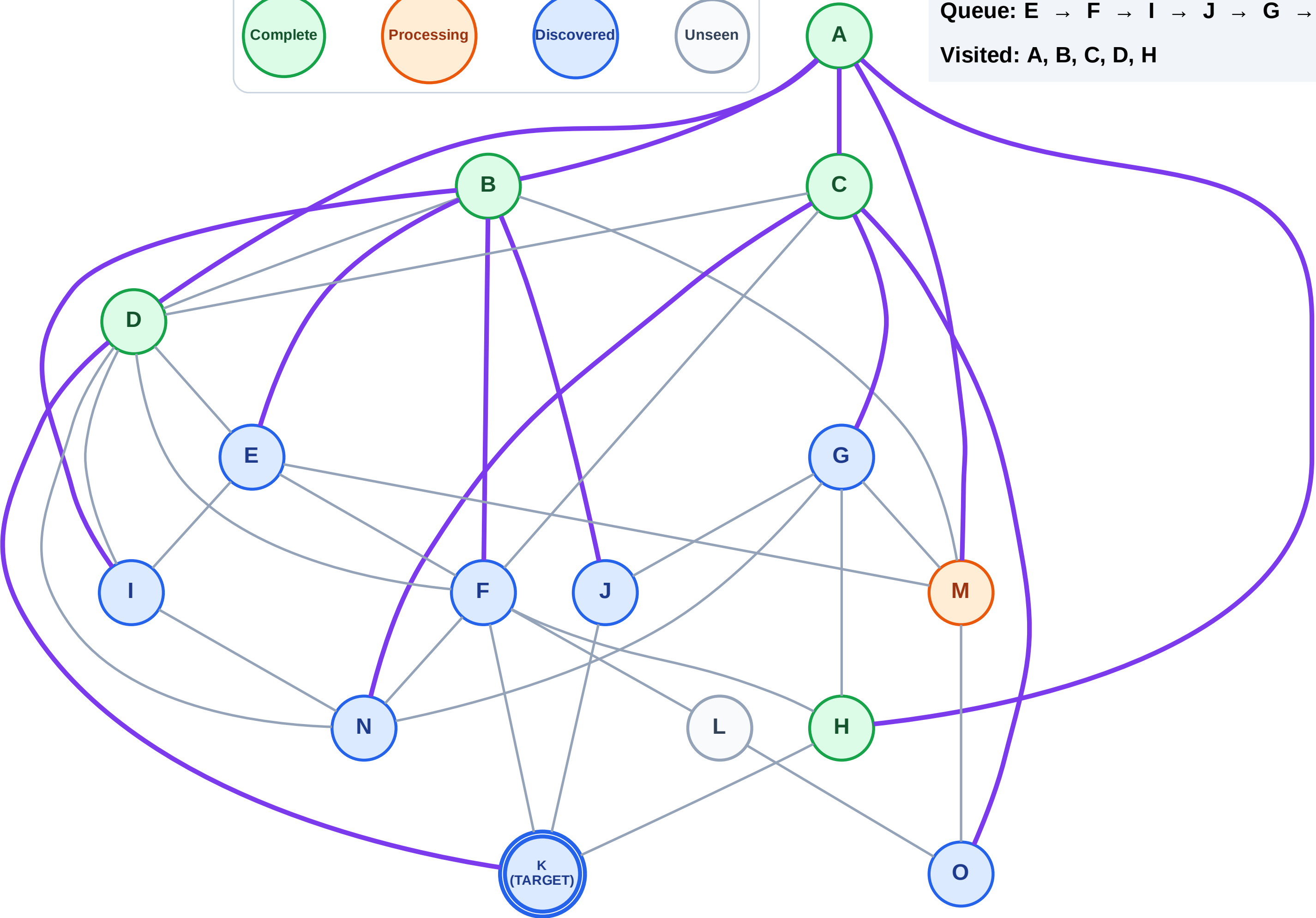
Processing

Discovered

Unseen

Queue: E → F → I → J → G → N → O → K

Visited: A, B, C, D, H



BFS Traversal

Step 13: No new neighbors from M

Legend

Complete

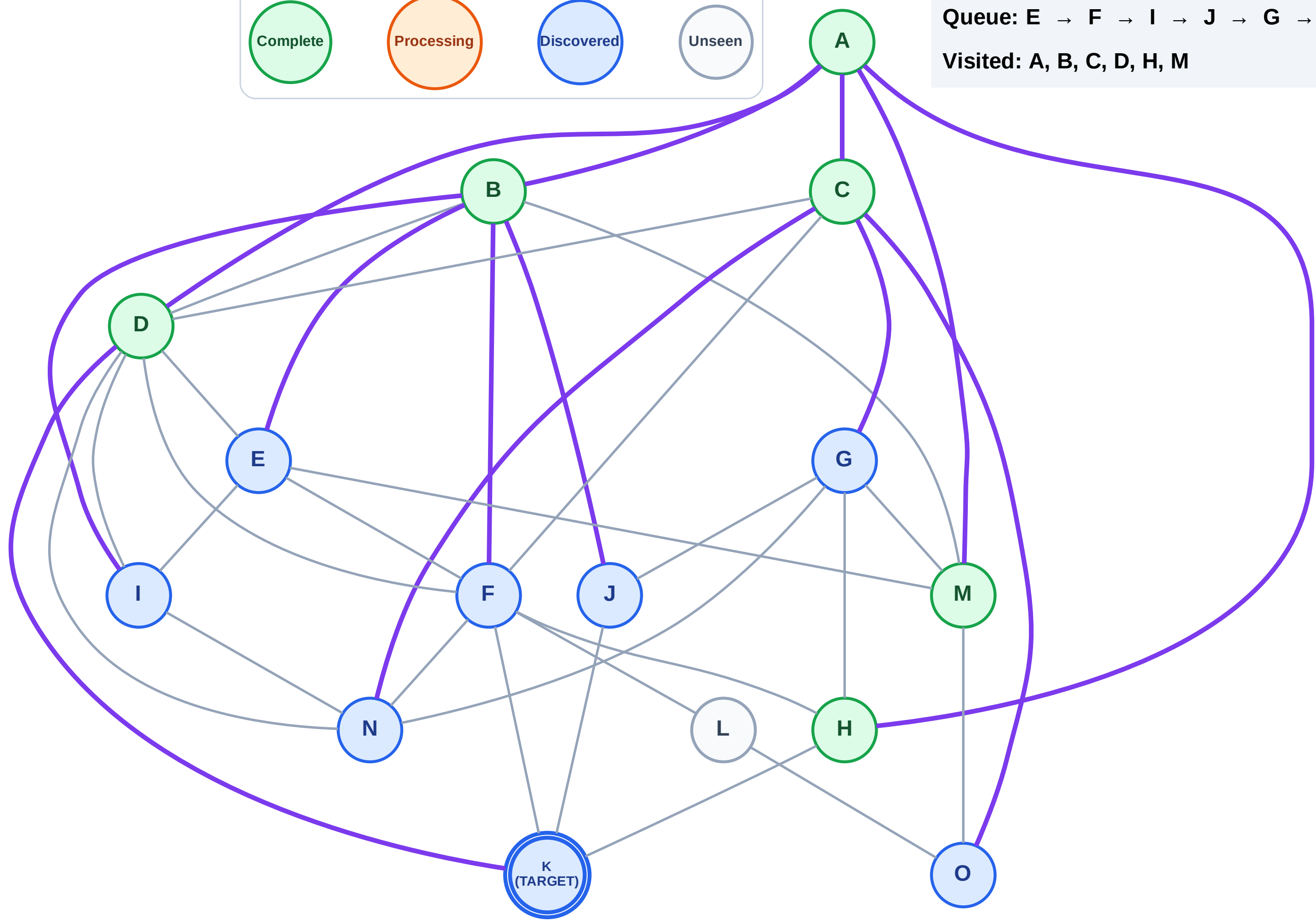
Processing

Discovered

Unseen

Queue: E → F → I → J → G → N → O → K

Visited: A, B, C, D, H, M



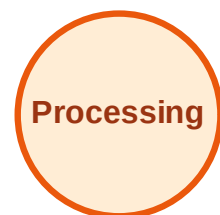
BFS Traversal

Step 14: Process node E

Legend



Complete



Processing



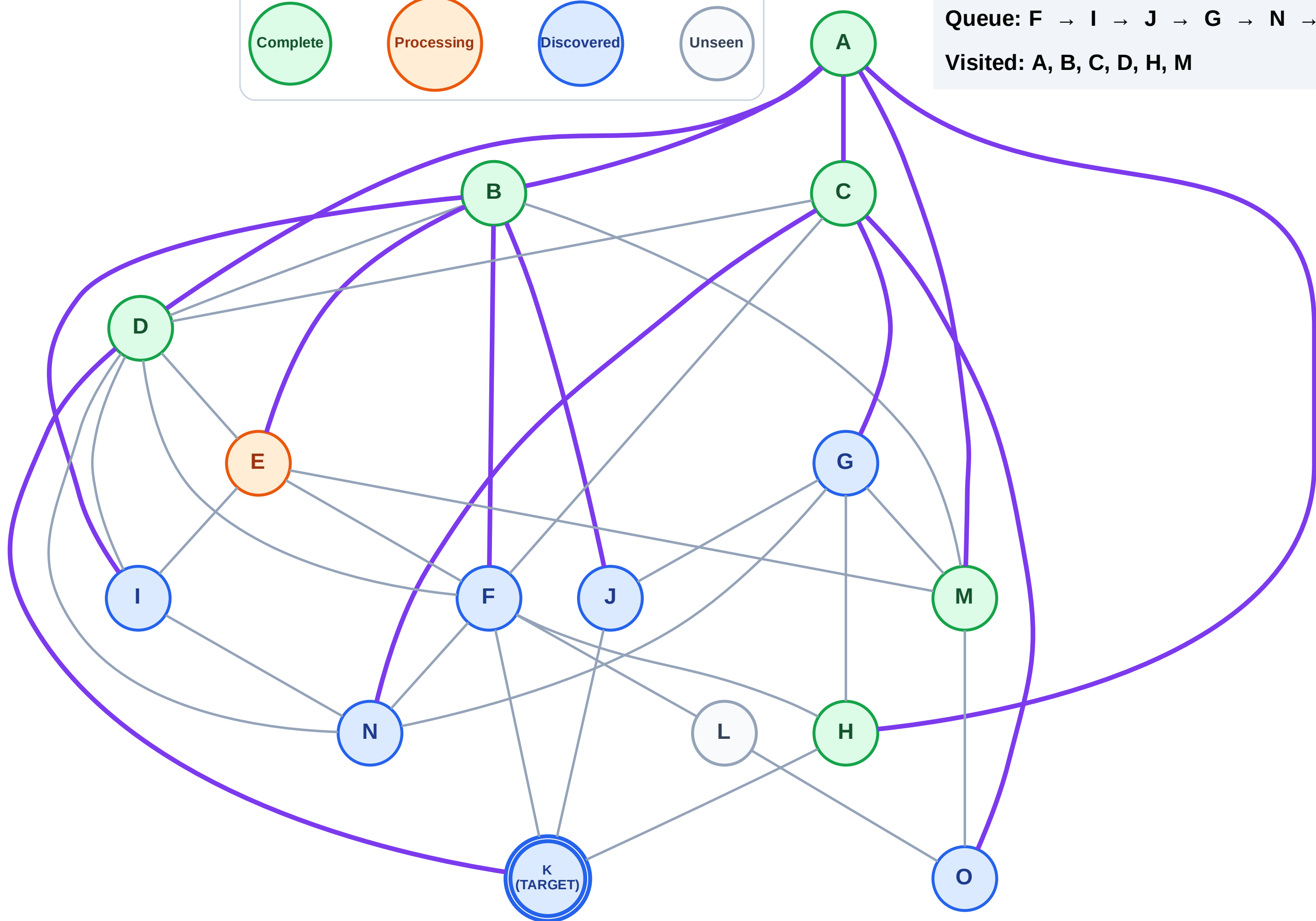
Discovered



Unseen

Queue: F → I → J → G → N → O → K

Visited: A, B, C, D, H, M



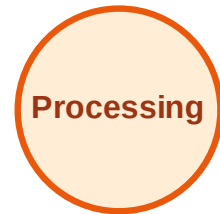
BFS Traversal

Step 15: No new neighbors from E

Legend



Complete



Processing



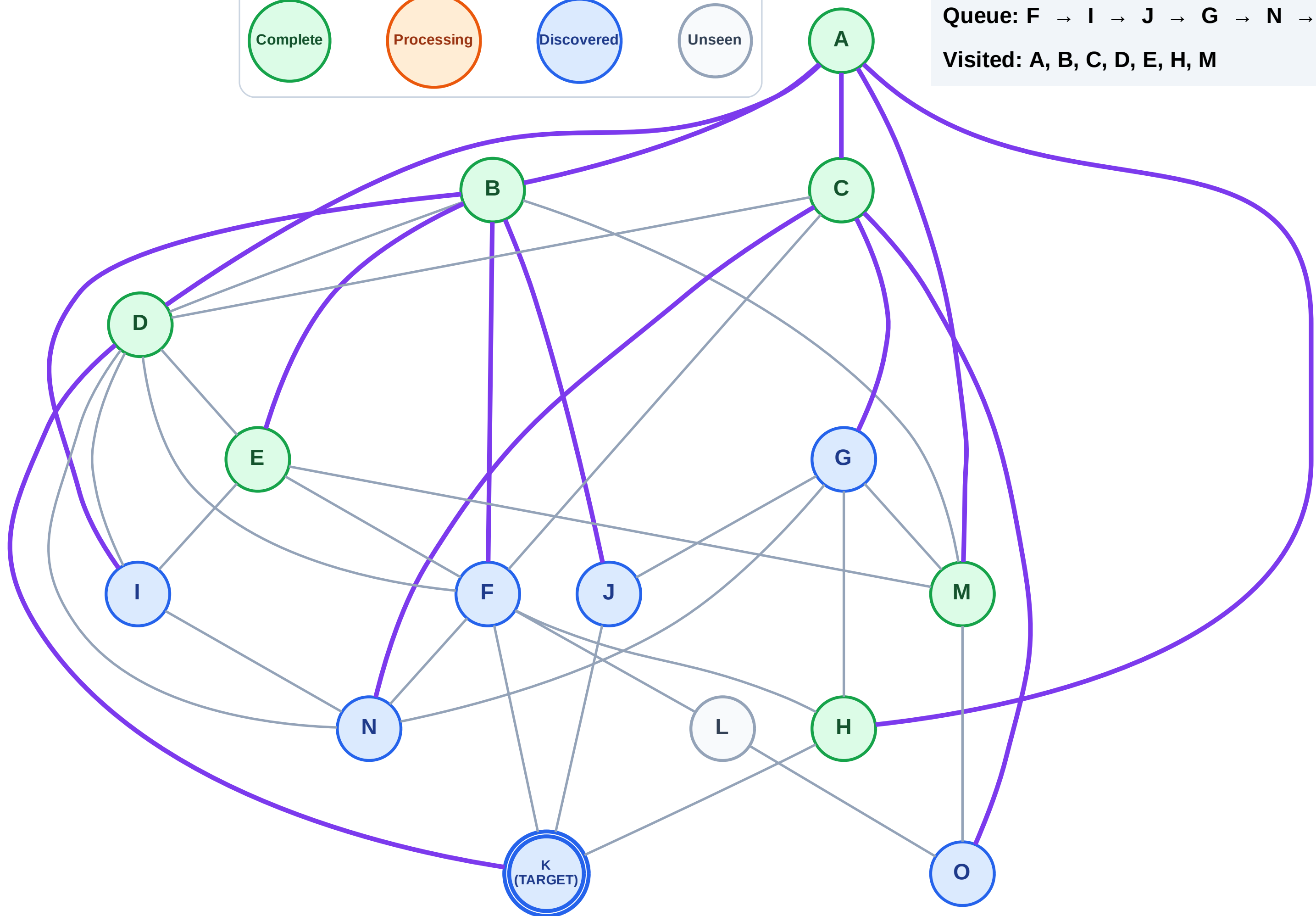
Discovered



Unseen

Queue: F → I → J → G → N → O → K

Visited: A, B, C, D, E, H, M



BFS Traversal

Step 16: Process node F

Legend

Complete

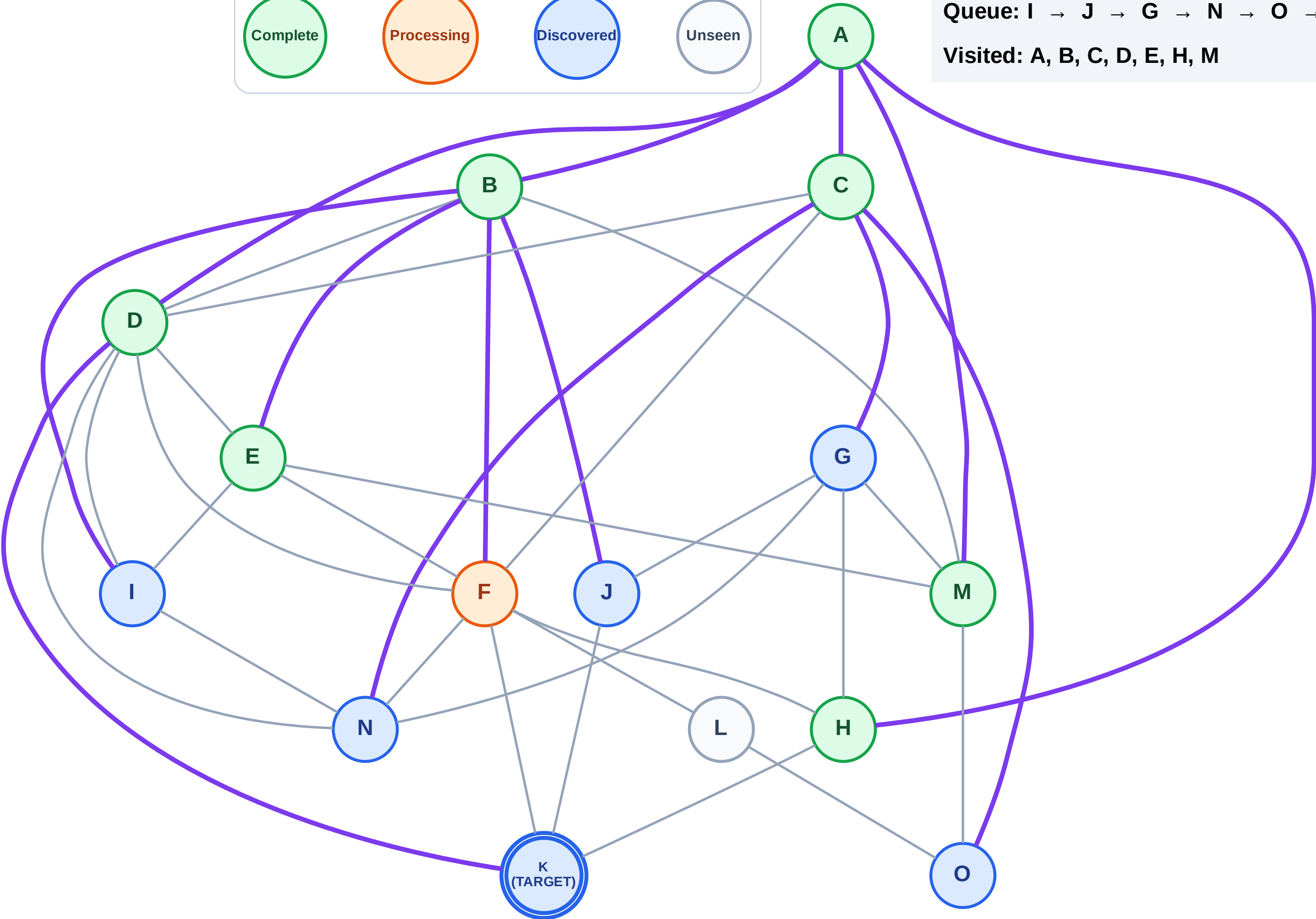
Processing

Discovered

Unseen

Queue: I → J → G → N → O → K

Visited: A, B, C, D, E, H, M



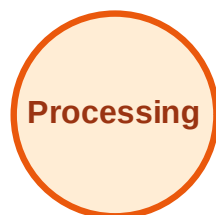
BFS Traversal

Step 17: Discover L from F

Legend



Complete



Processing



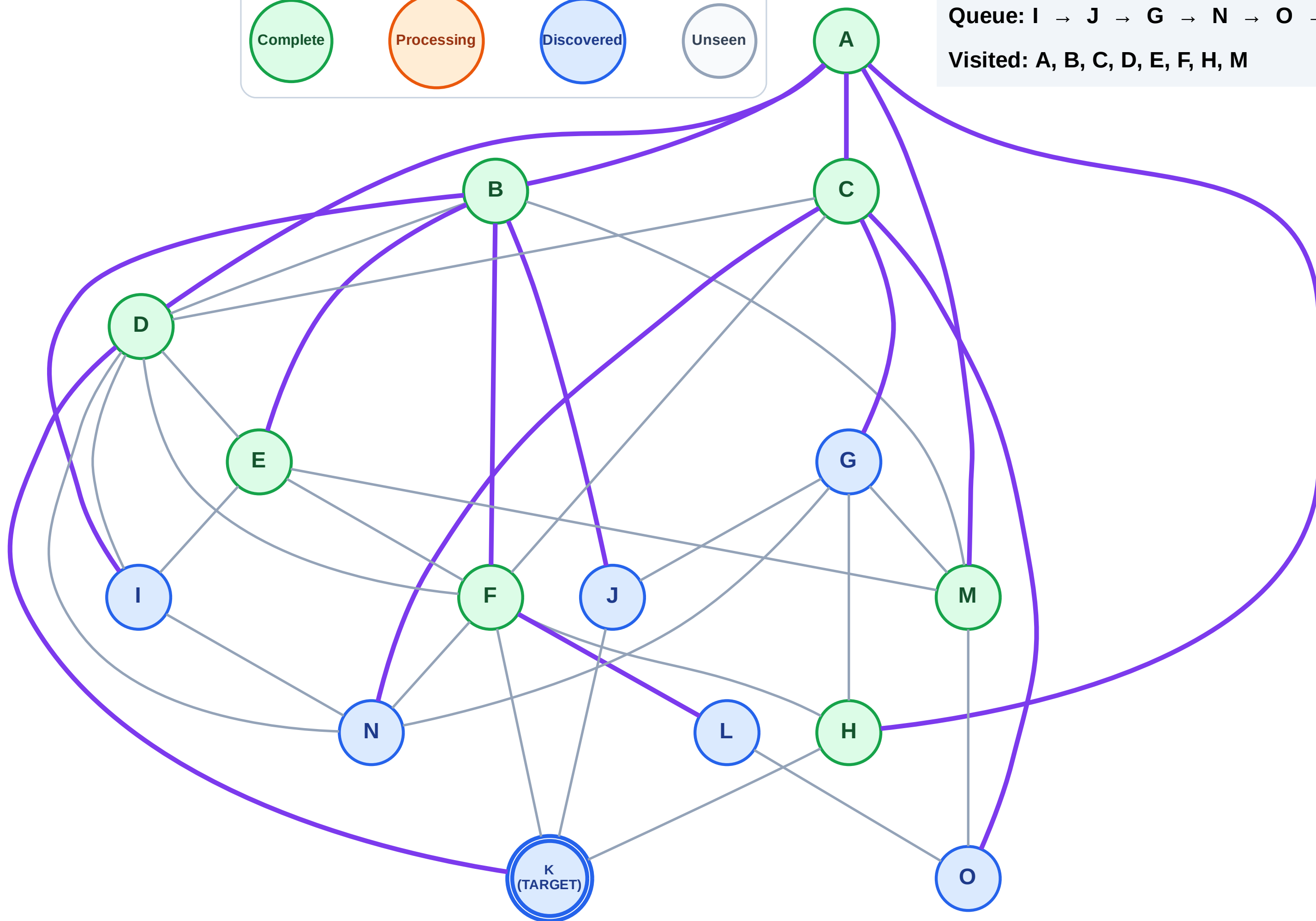
Discovered



Unseen

Queue: I → J → G → N → O → K → L

Visited: A, B, C, D, E, F, H, M



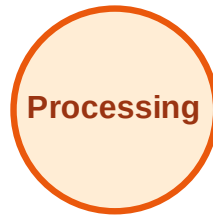
BFS Traversal

Step 18: Process node I

Legend



Complete



Processing



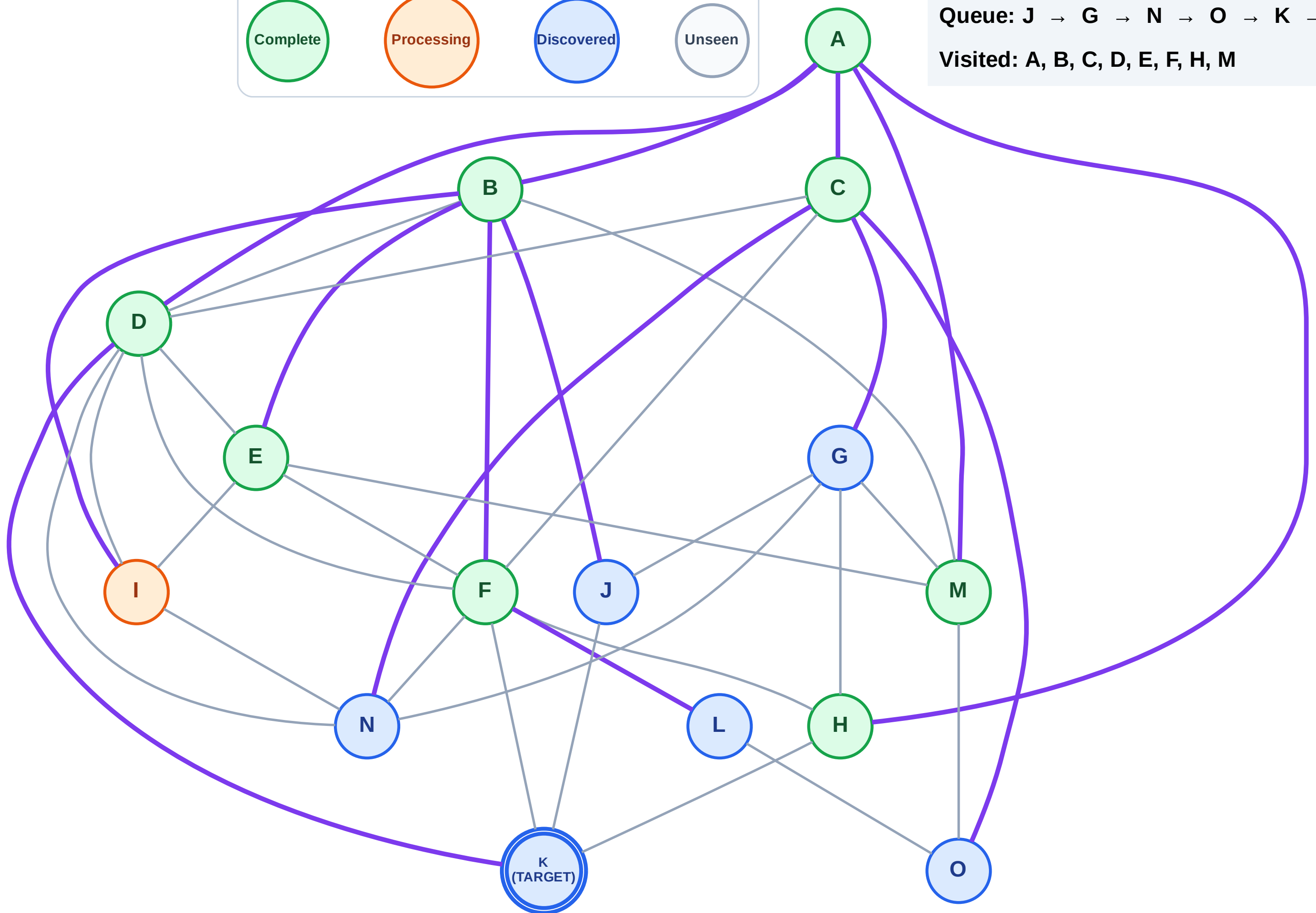
Discovered



Unseen

Queue: J → G → N → O → K → L

Visited: A, B, C, D, E, F, H, M



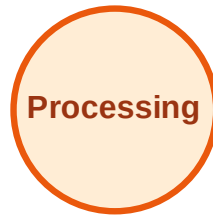
BFS Traversal

Step 19: No new neighbors from I

Legend



Complete



Processing



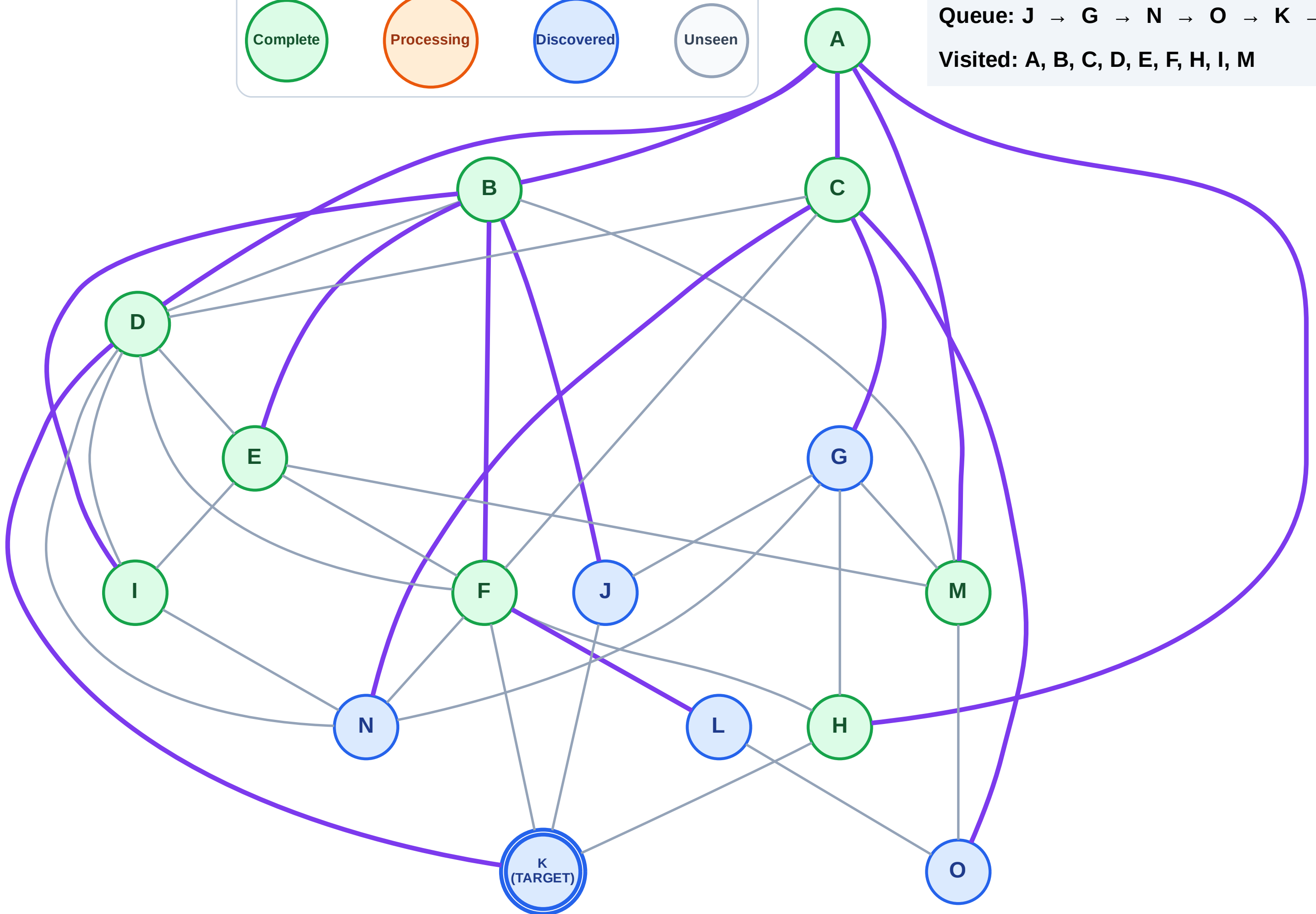
Discovered



Unseen

Queue: J → G → N → O → K → L

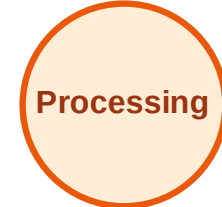
Visited: A, B, C, D, E, F, H, I, M



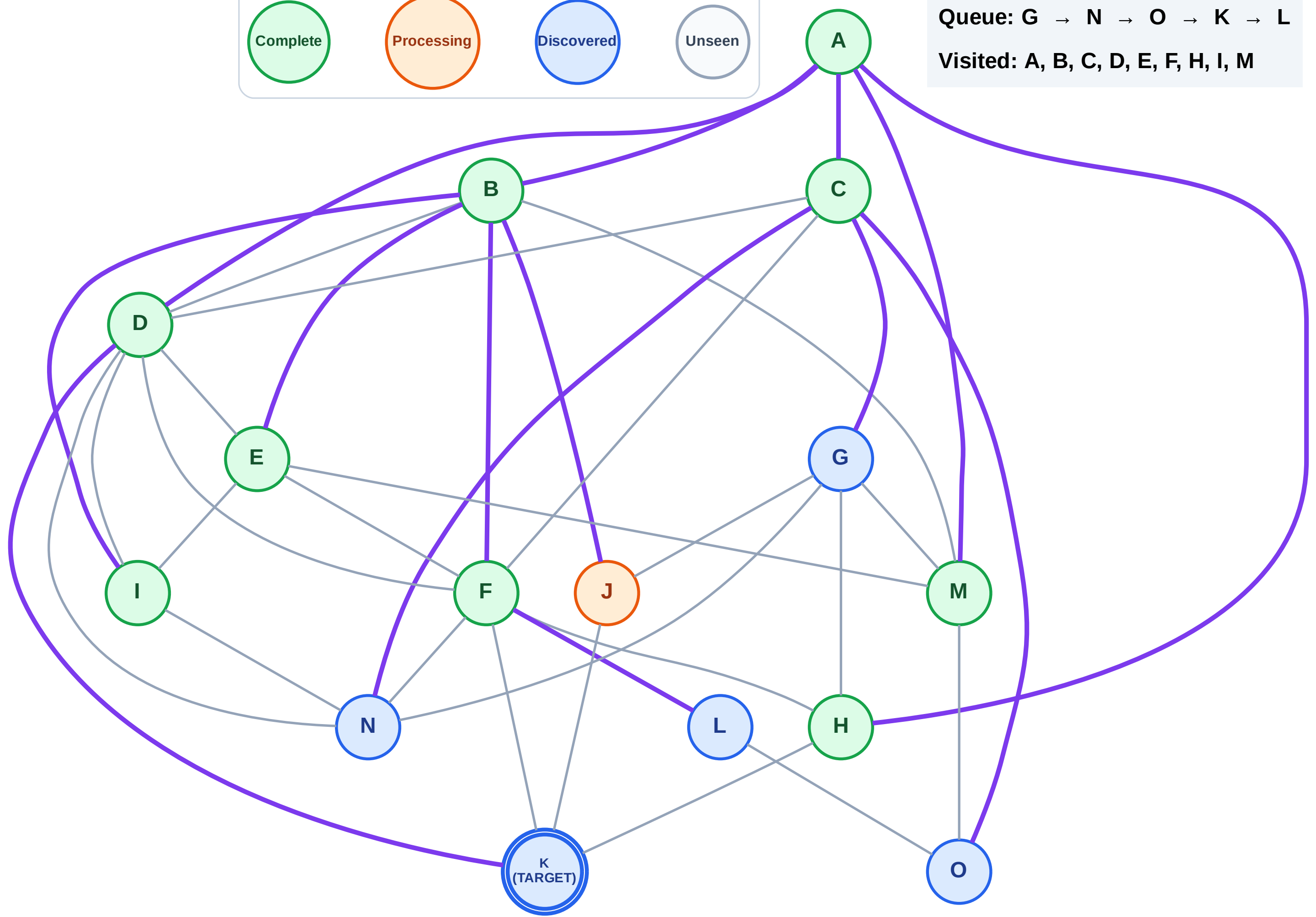
BFS Traversal

Step 20: Process node J

Legend



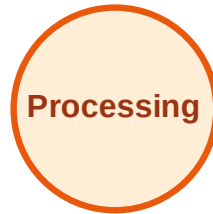
Queue: G → N → O → K → L
Visited: A, B, C, D, E, F, H, I, M



BFS Traversal

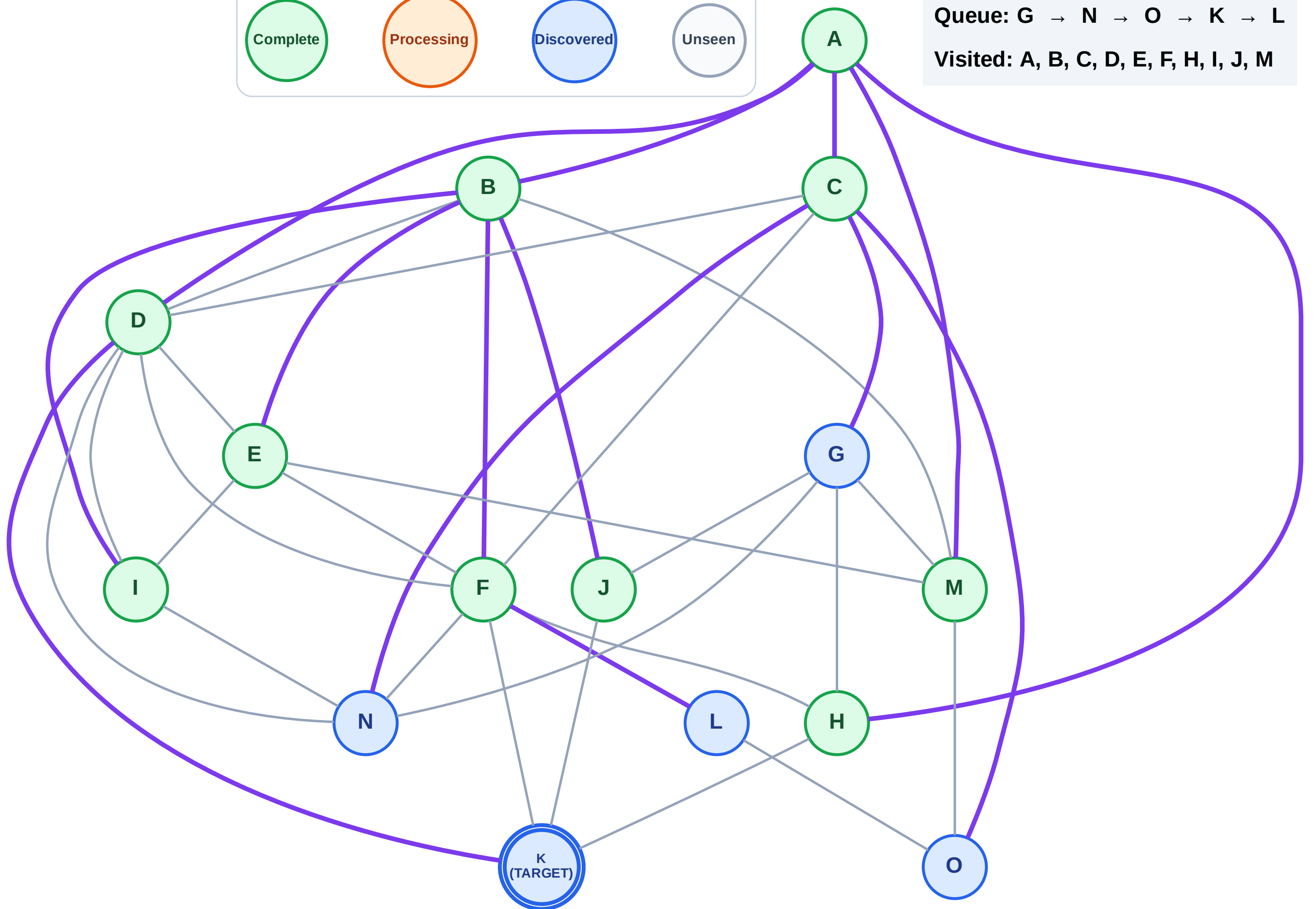
Step 21: No new neighbors from J

Legend



Queue: G → N → O → K → L

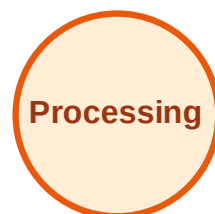
Visited: A, B, C, D, E, F, H, I, J, M



BFS Traversal

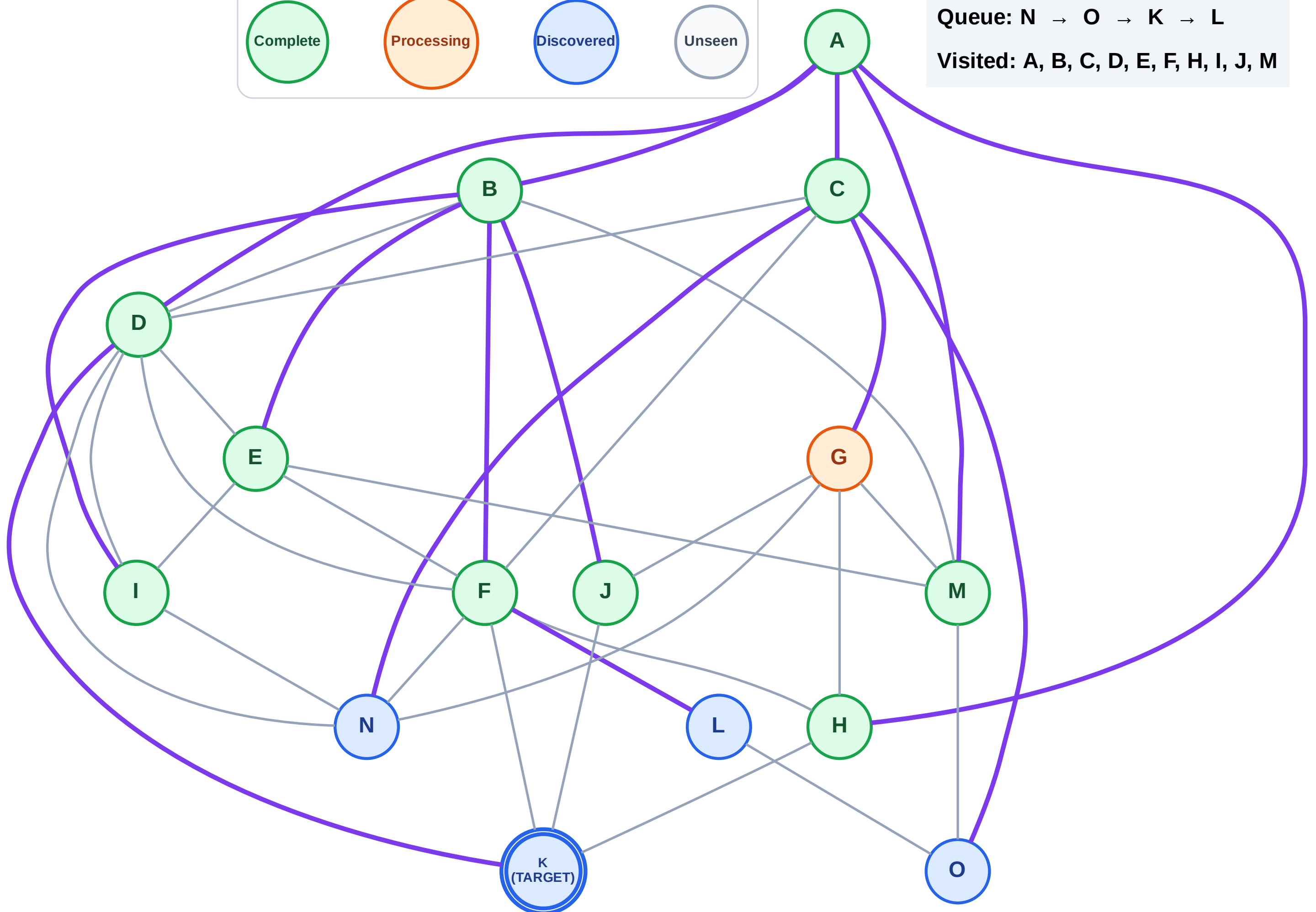
Step 22: Process node G

Legend



Queue: N → O → K → L

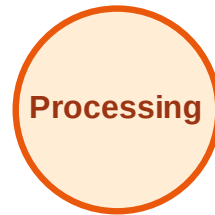
Visited: A, B, C, D, E, F, H, I, J, M



BFS Traversal

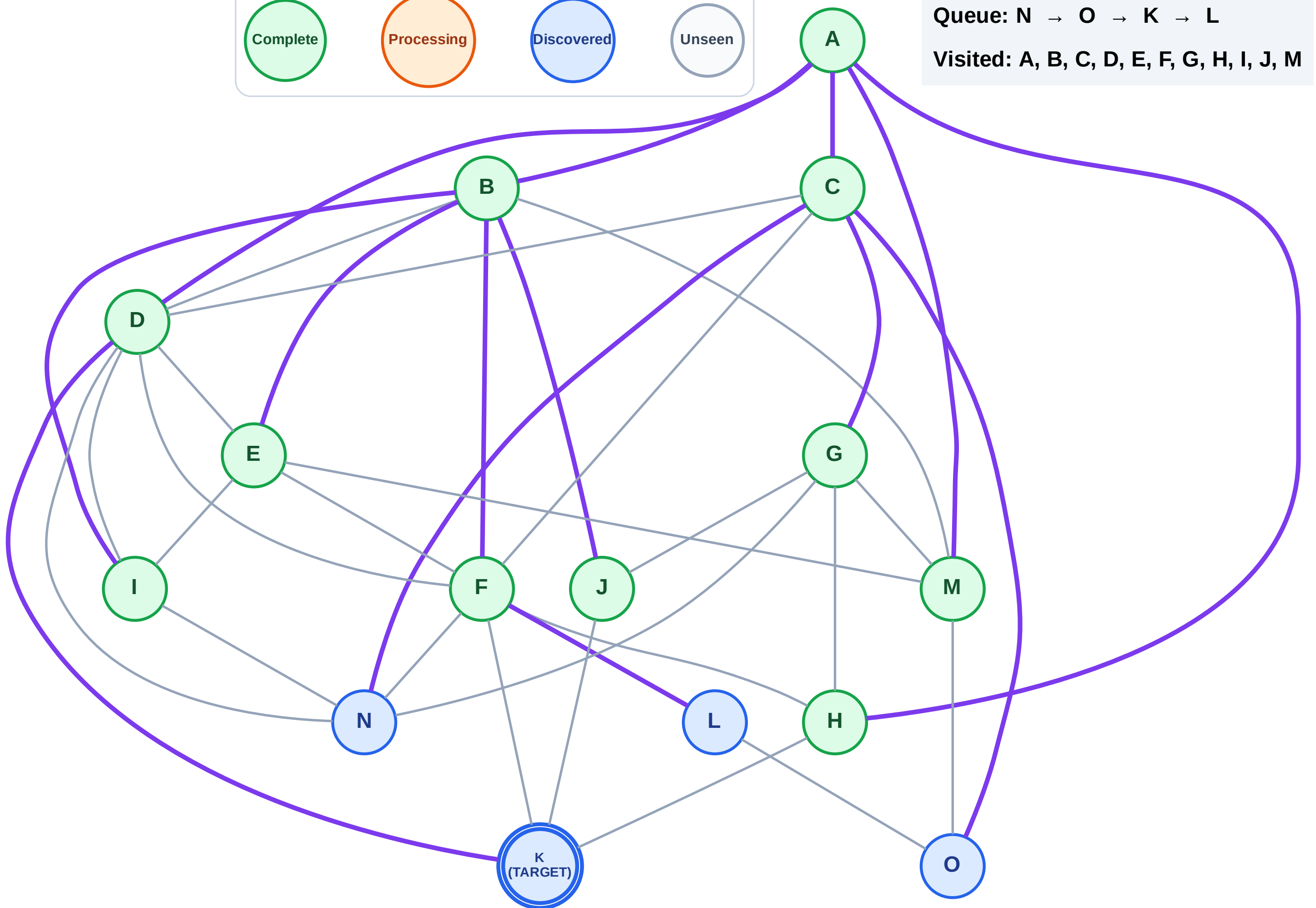
Step 23: No new neighbors from G

Legend



Queue: N → O → K → L

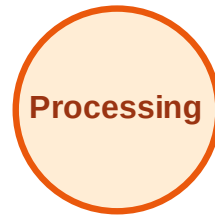
Visited: A, B, C, D, E, F, G, H, I, J, M



BFS Traversal

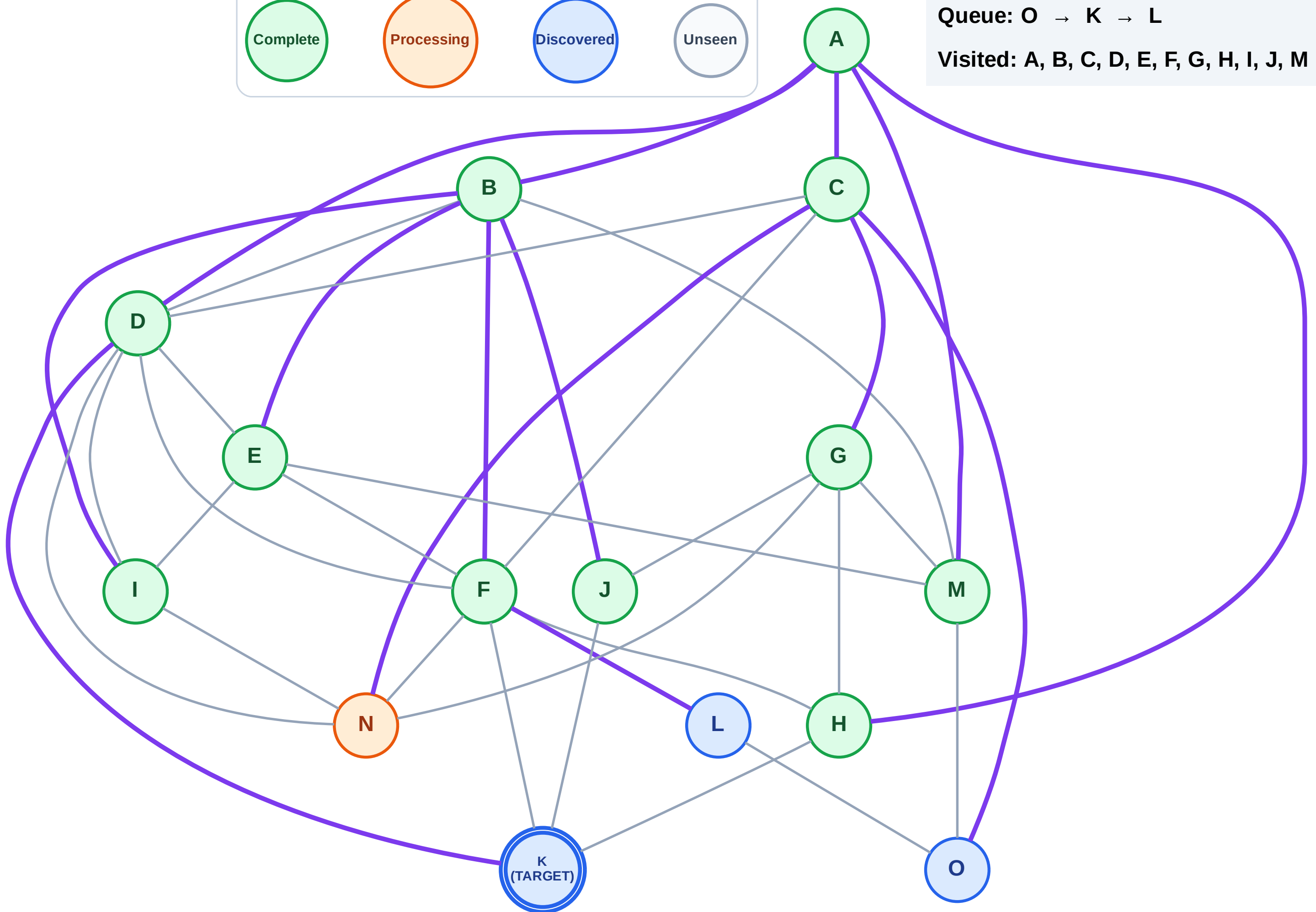
Step 24: Process node N

Legend



Queue: O → K → L

Visited: A, B, C, D, E, F, G, H, I, J, M



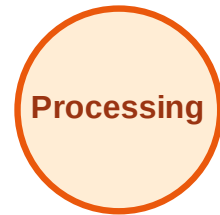
BFS Traversal

Step 25: No new neighbors from N

Legend



Complete



Processing



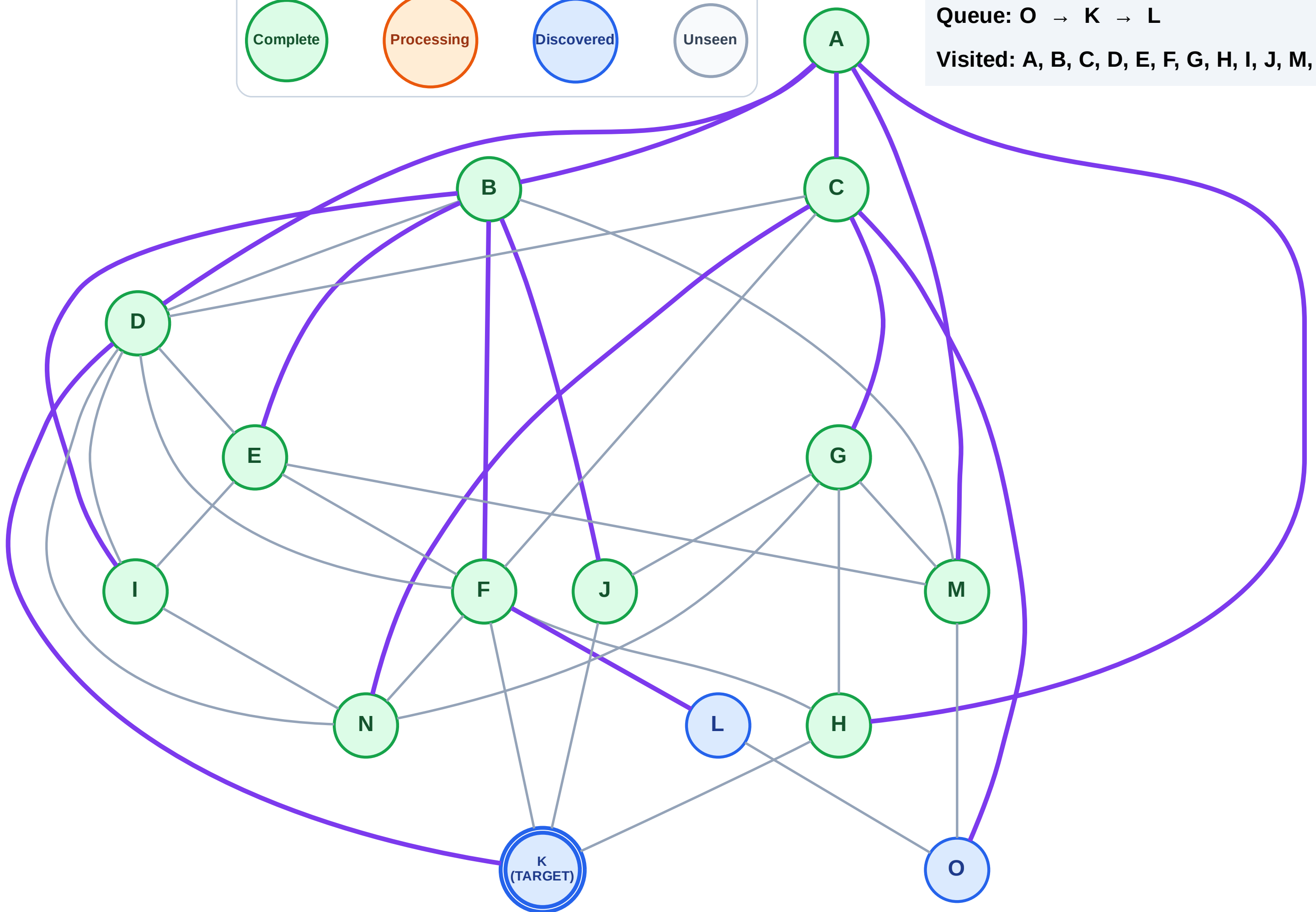
Discovered



Unseen

Queue: O → K → L

Visited: A, B, C, D, E, F, G, H, I, J, M, N



BFS Traversal

Step 26: Process node O

Legend

Complete

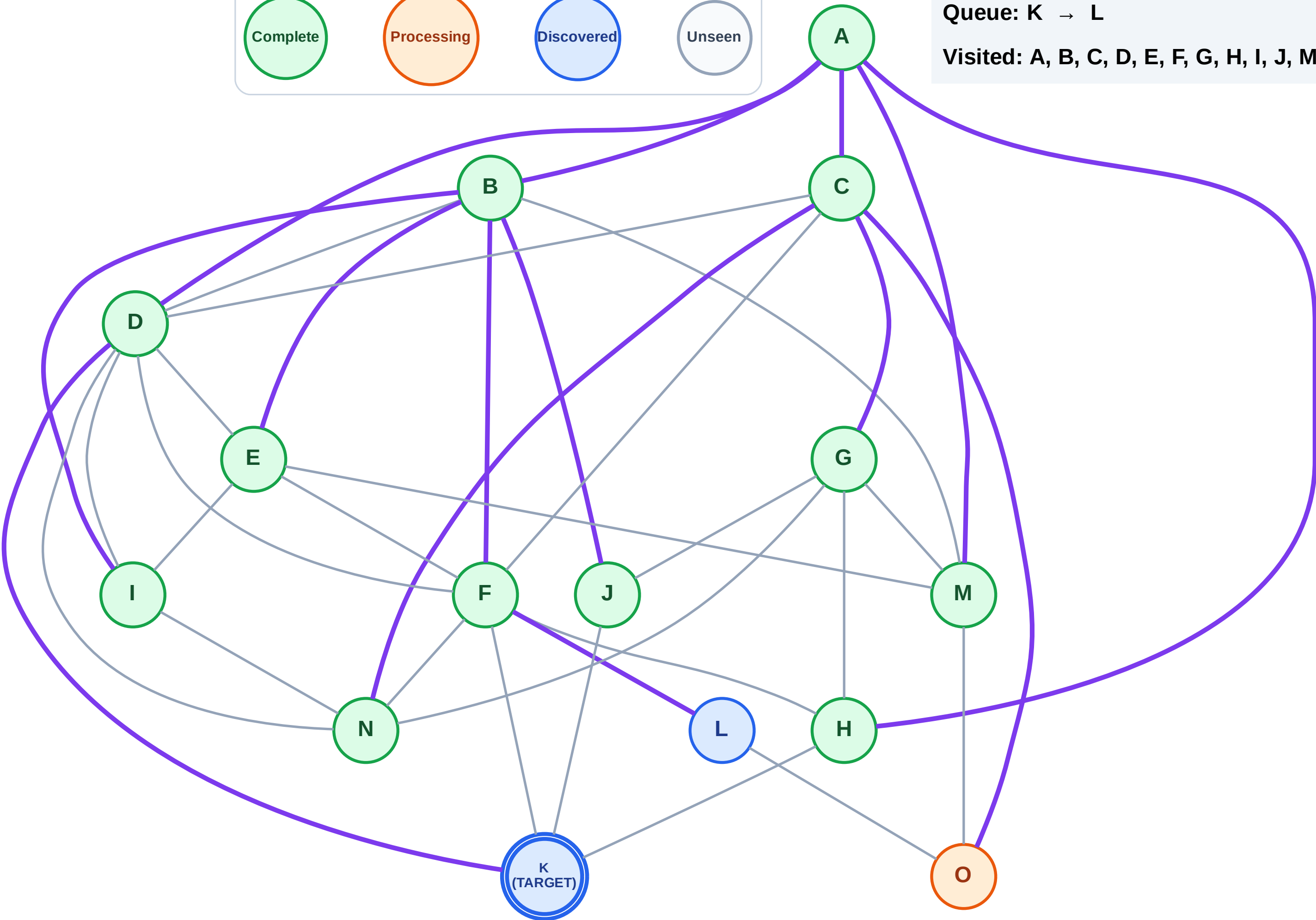
Processing

Discovered

Unseen

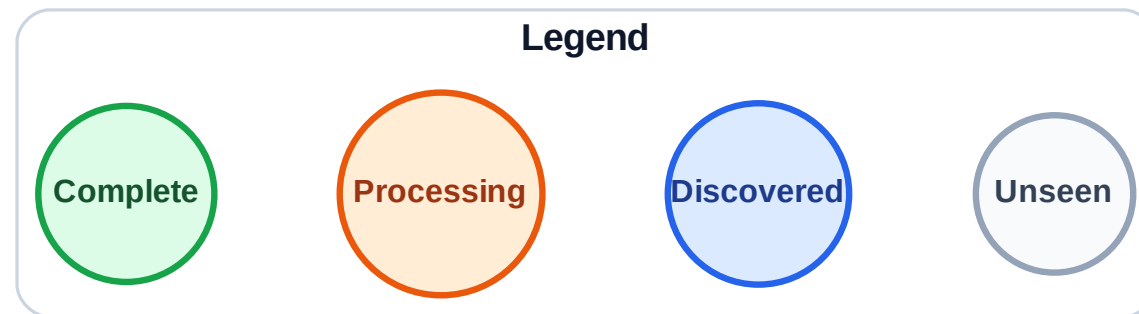
Queue: K → L

Visited: A, B, C, D, E, F, G, H, I, J, M, N



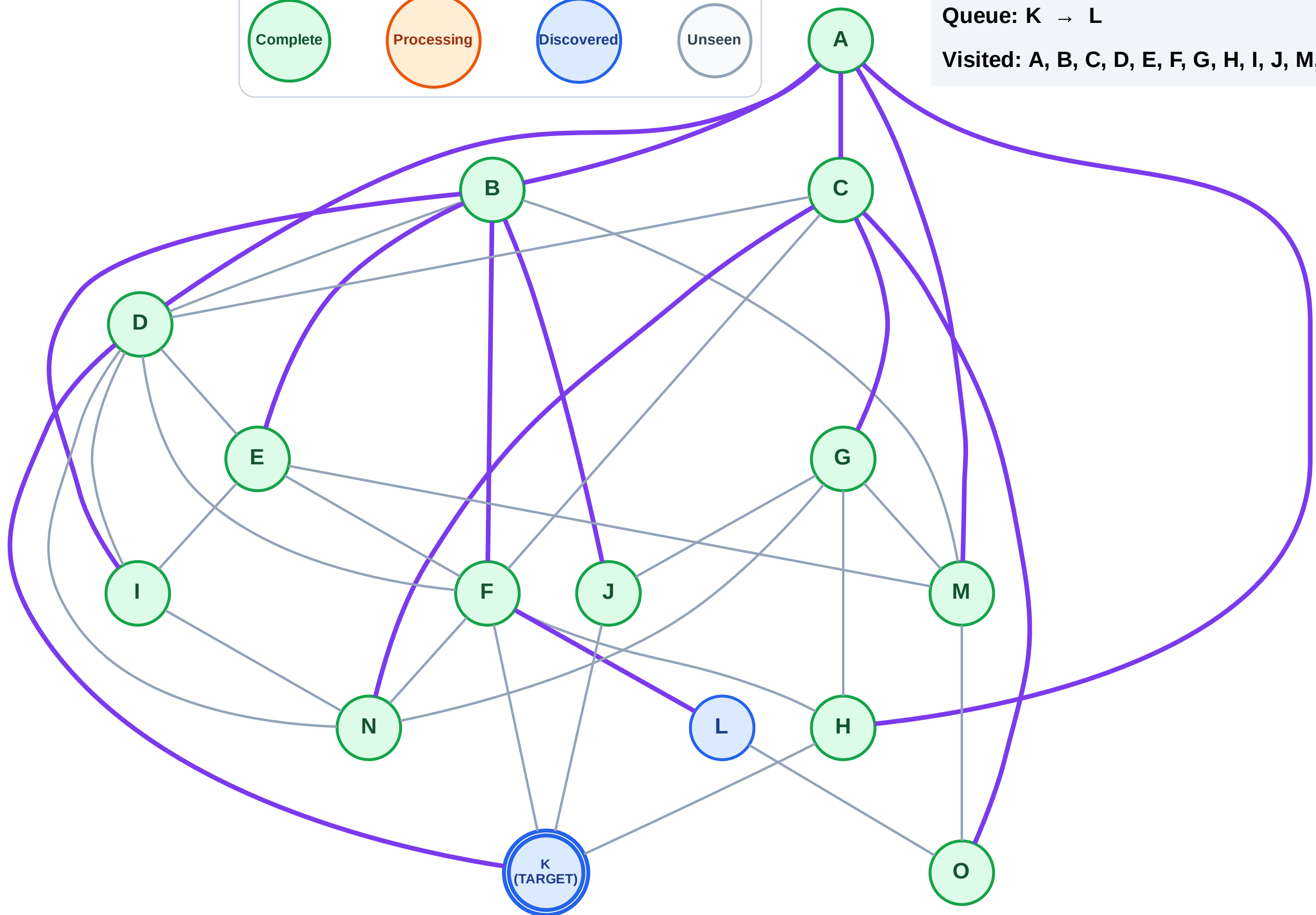
Step 27: No new neighbors from O

Step 27: No new neighbors from O



Queue: $K \rightarrow L$

Visited: A, B, C, D, E, F, G, H, I, J, M, N, O

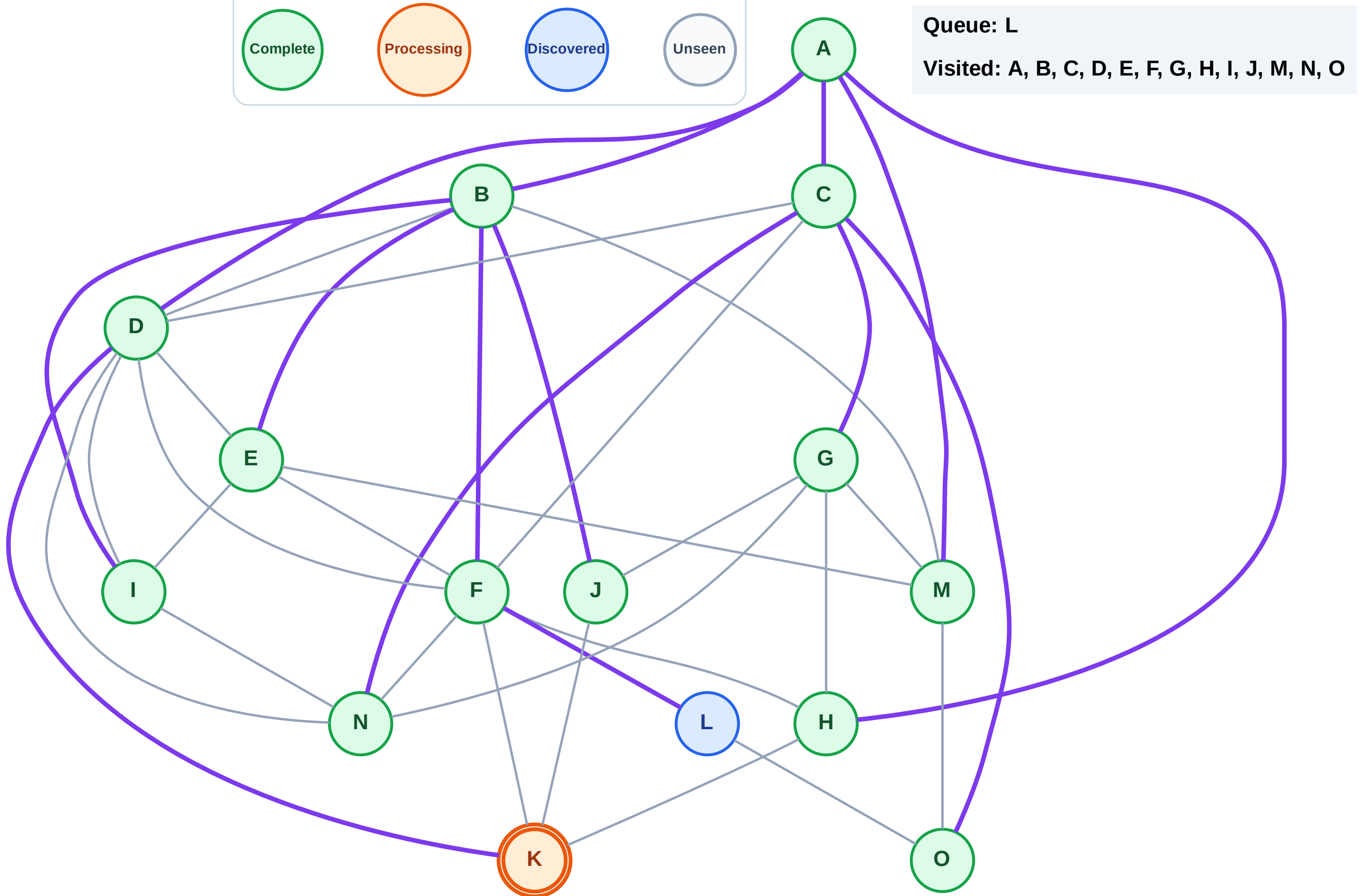


Step 28: Process node K

Page 10 of 10



Visited: A



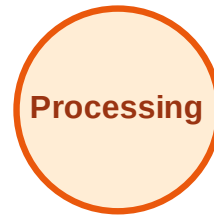
BFS Traversal

Step 29: TARGET NODE K REACHED! Path (2 hops): A → D → K

Legend



Complete



Processing



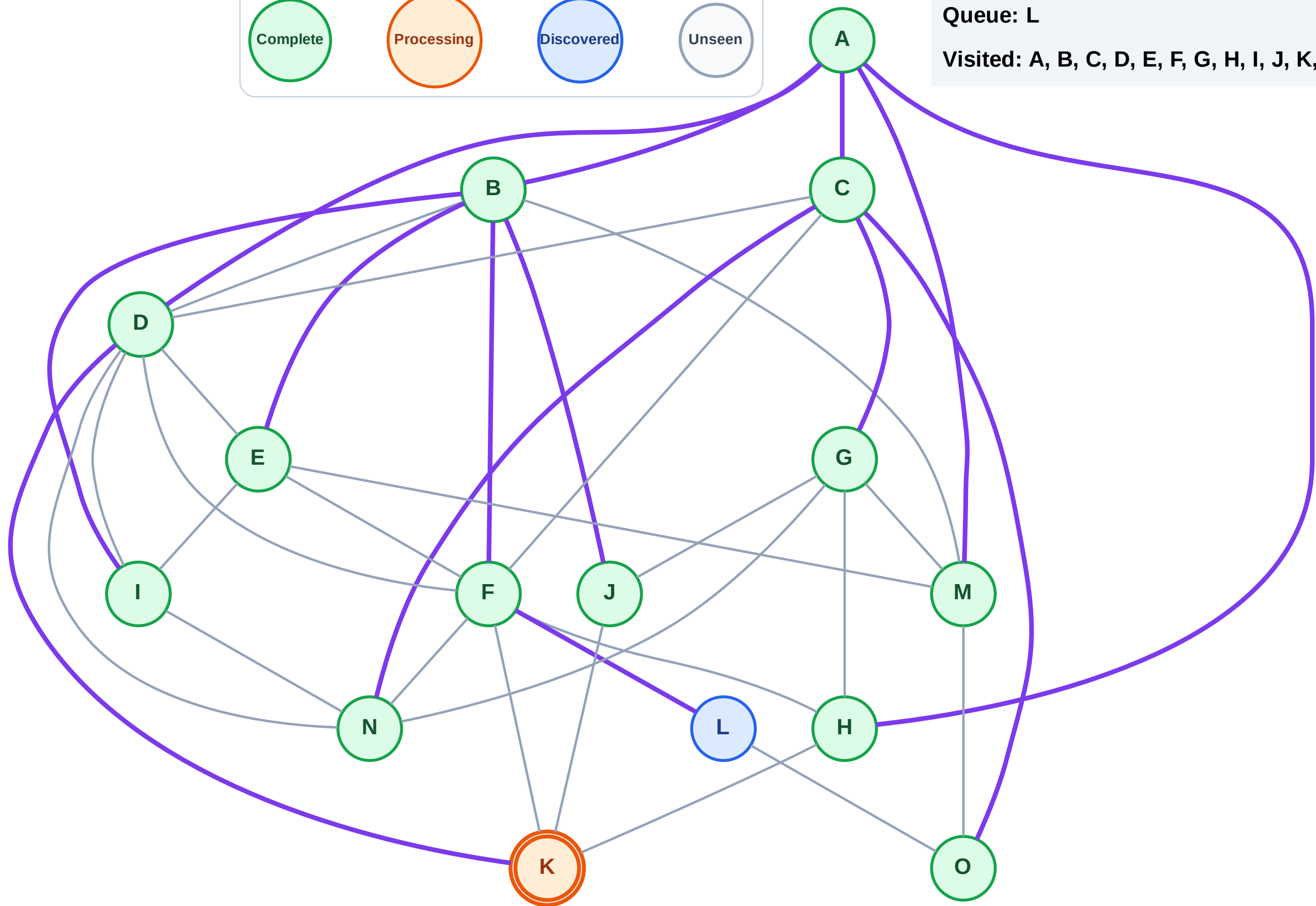
Discovered



Unseen

Queue: L

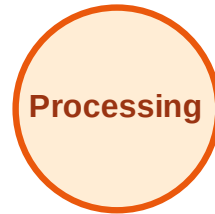
Visited: A, B, C, D, E, F, G, H, I, J, K, M, N, O



DFS Traversal

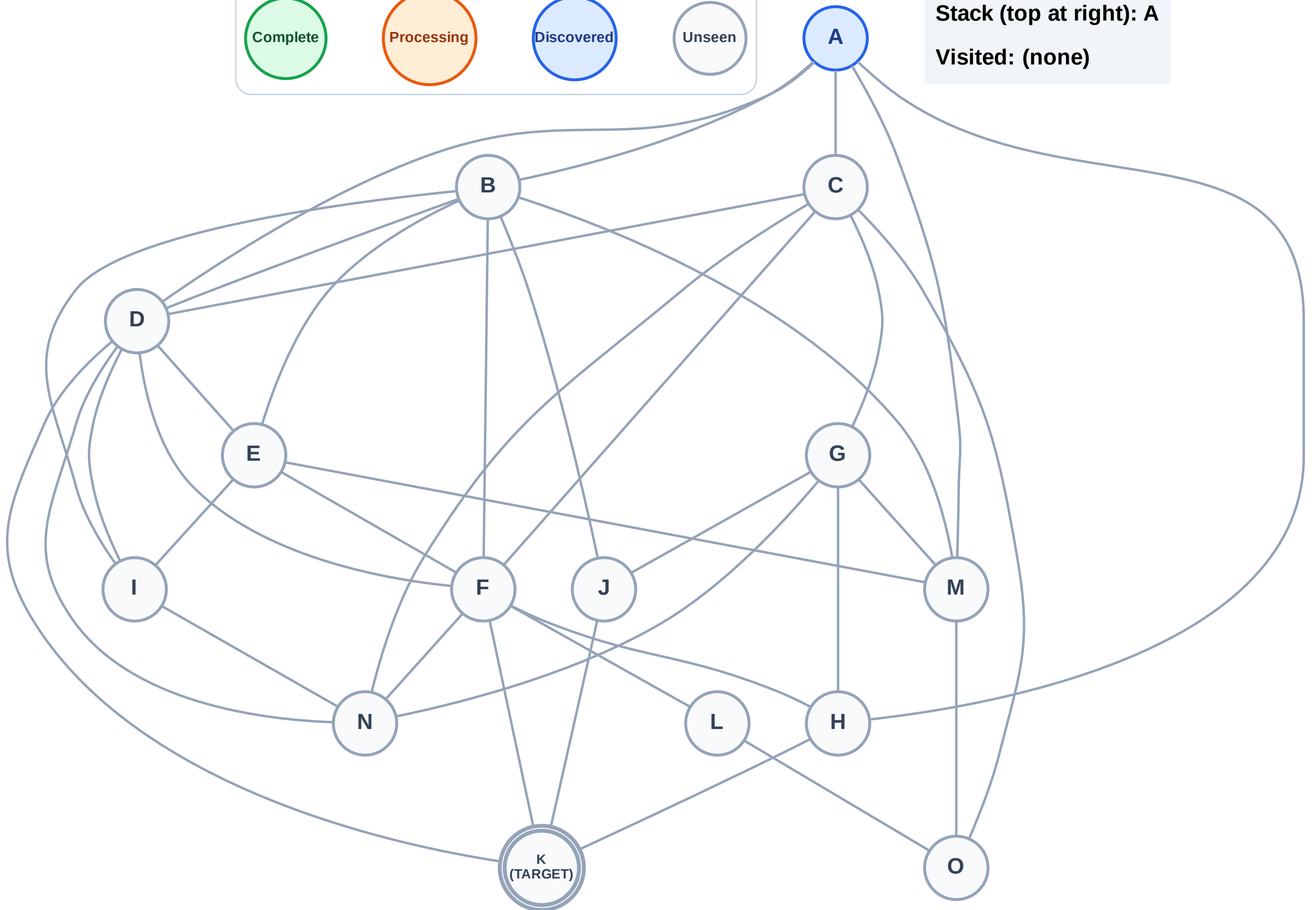
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

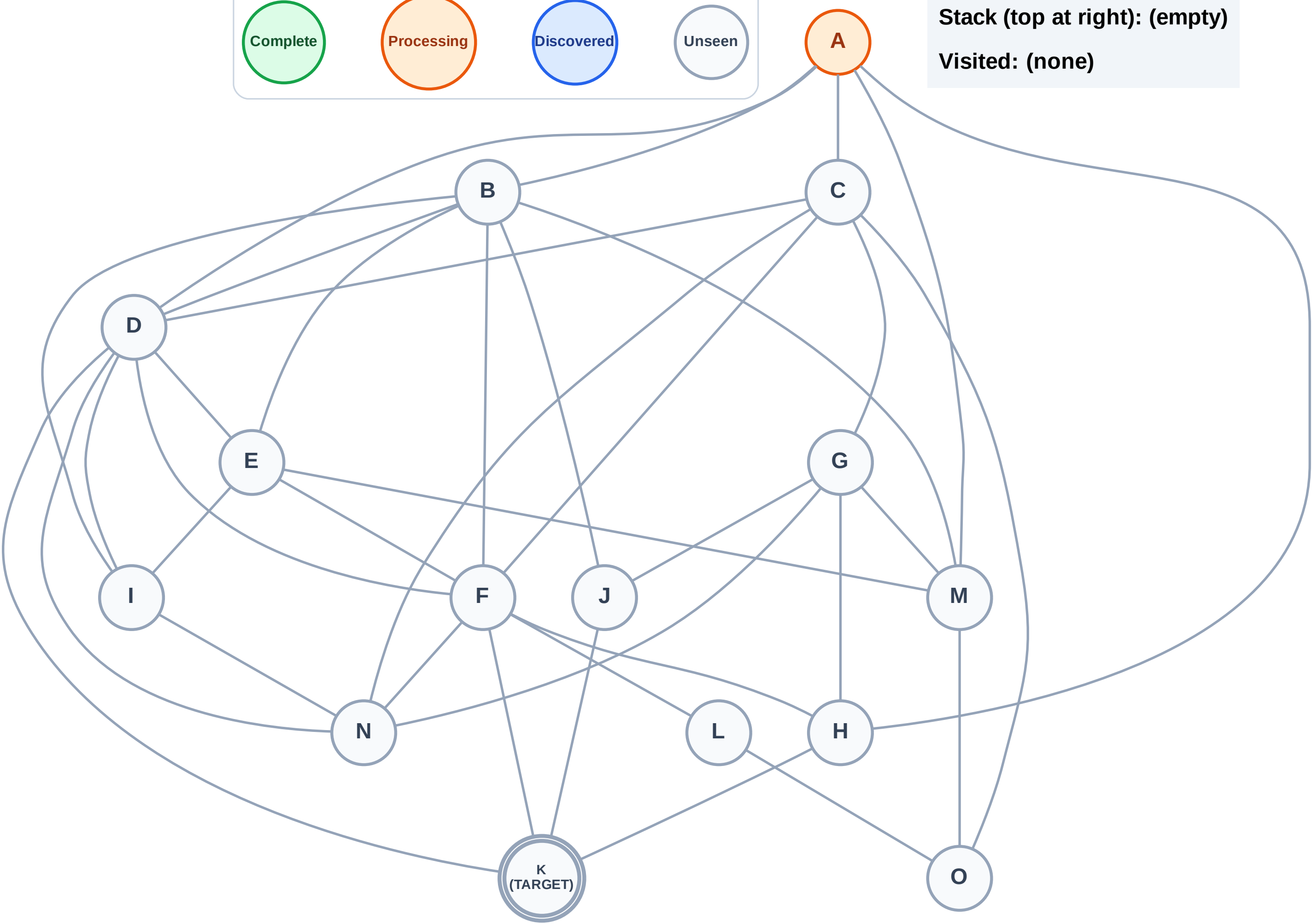
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



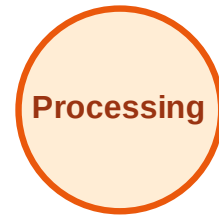
DFS Traversal

Step 3: Discover M, H, D, C, B from A

Legend



Complete



Processing



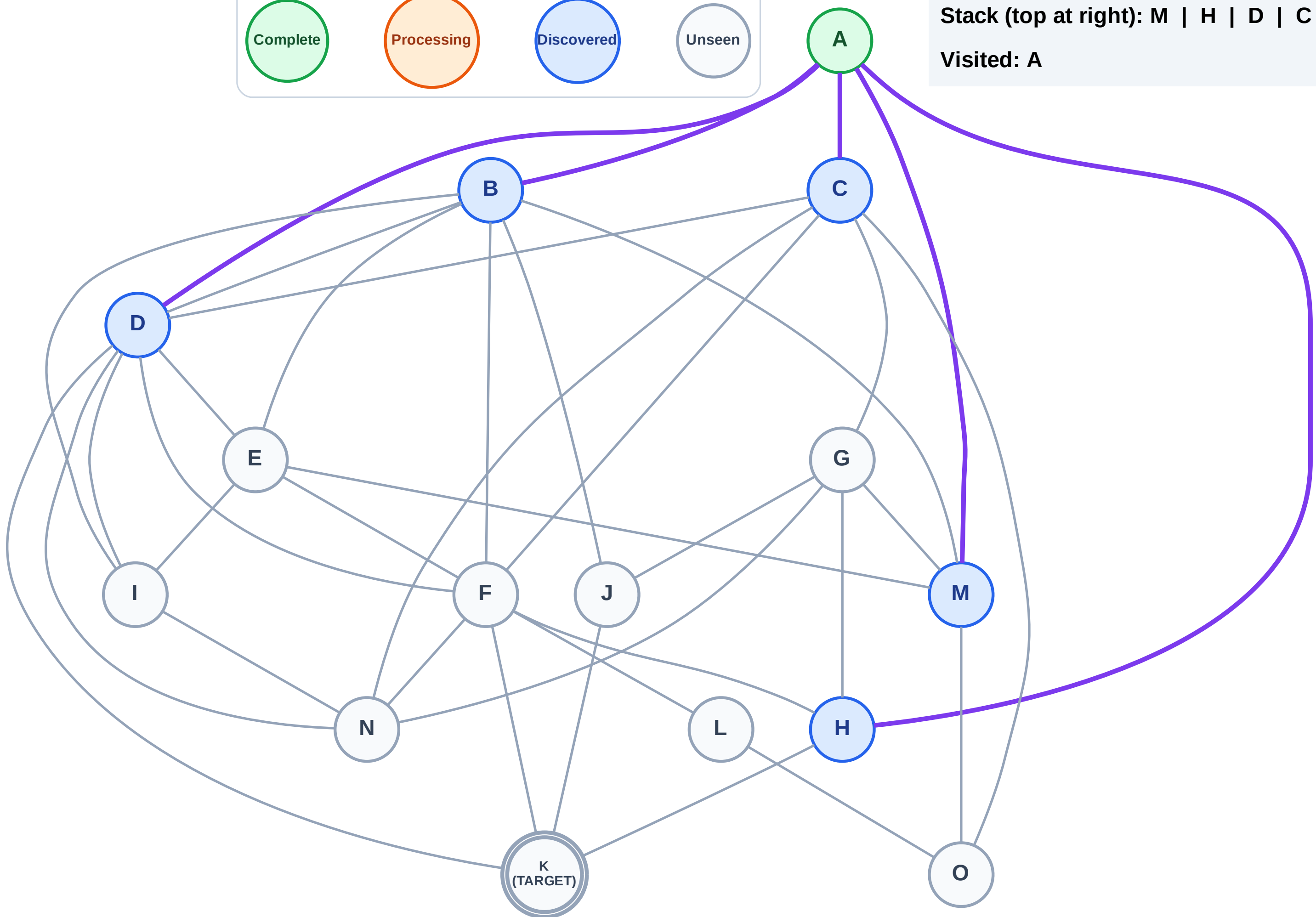
Discovered



Unseen

Stack (top at right): M | H | D | C | B

Visited: A



DFS Traversal

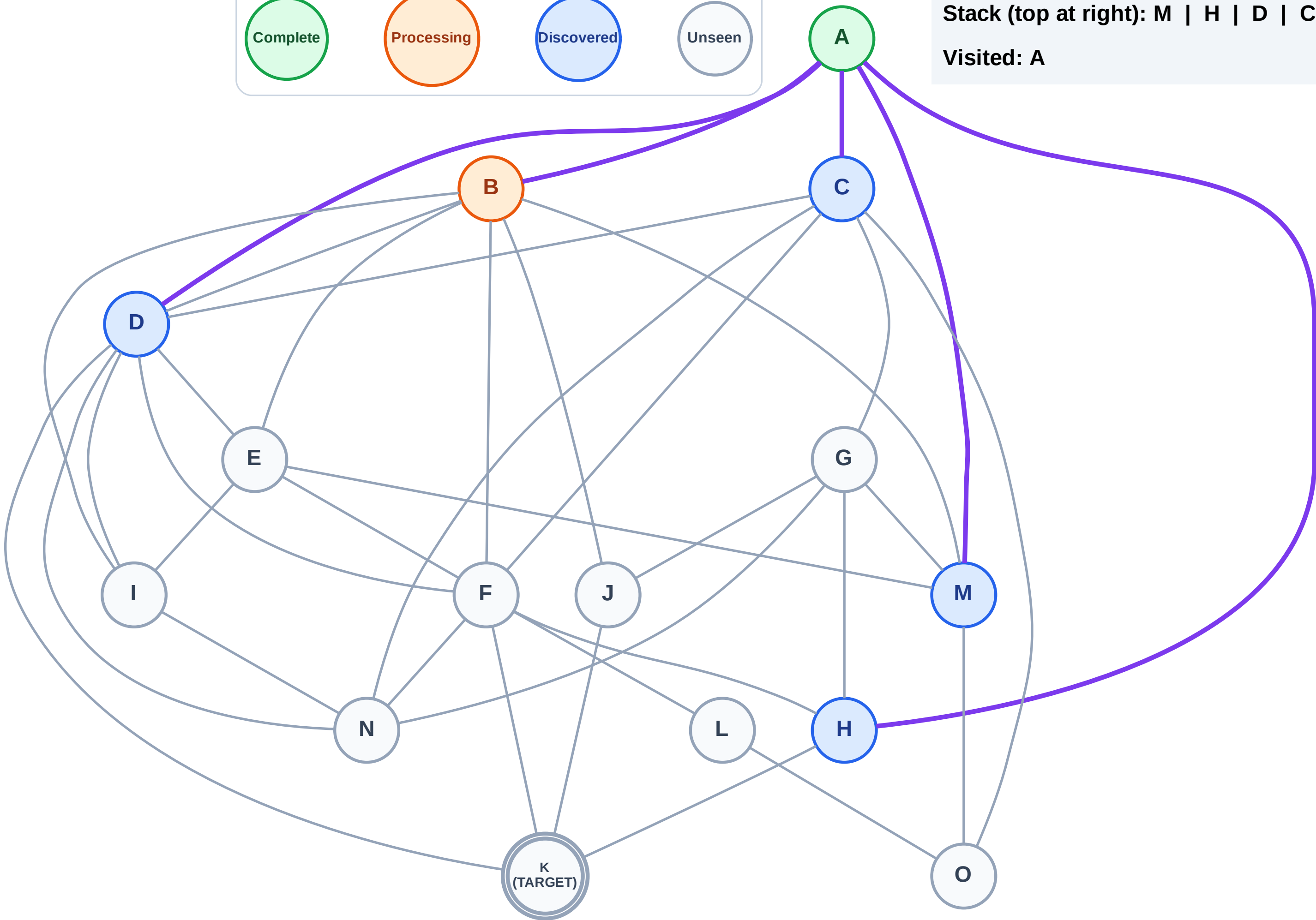
Step 4: Process node B

Legend



Stack (top at right): M | H | D | C

Visited: A



DFS Traversal

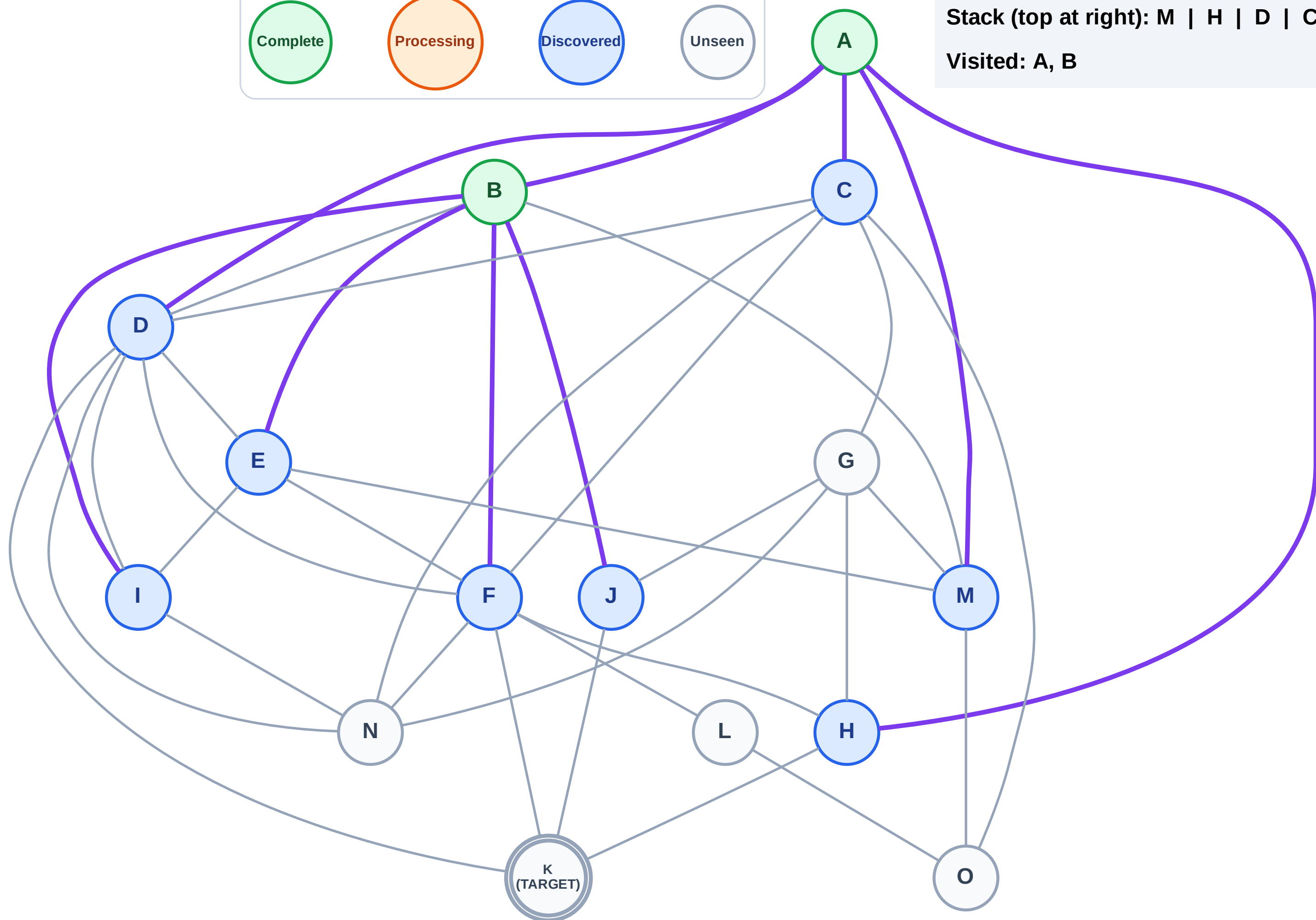
Step 5: Discover J, I, F, E from B

Legend



Stack (top at right): M | H | D | C | J | I | F | E

Visited: A, B



DFS Traversal

Step 6: Process node E

Legend

Complete

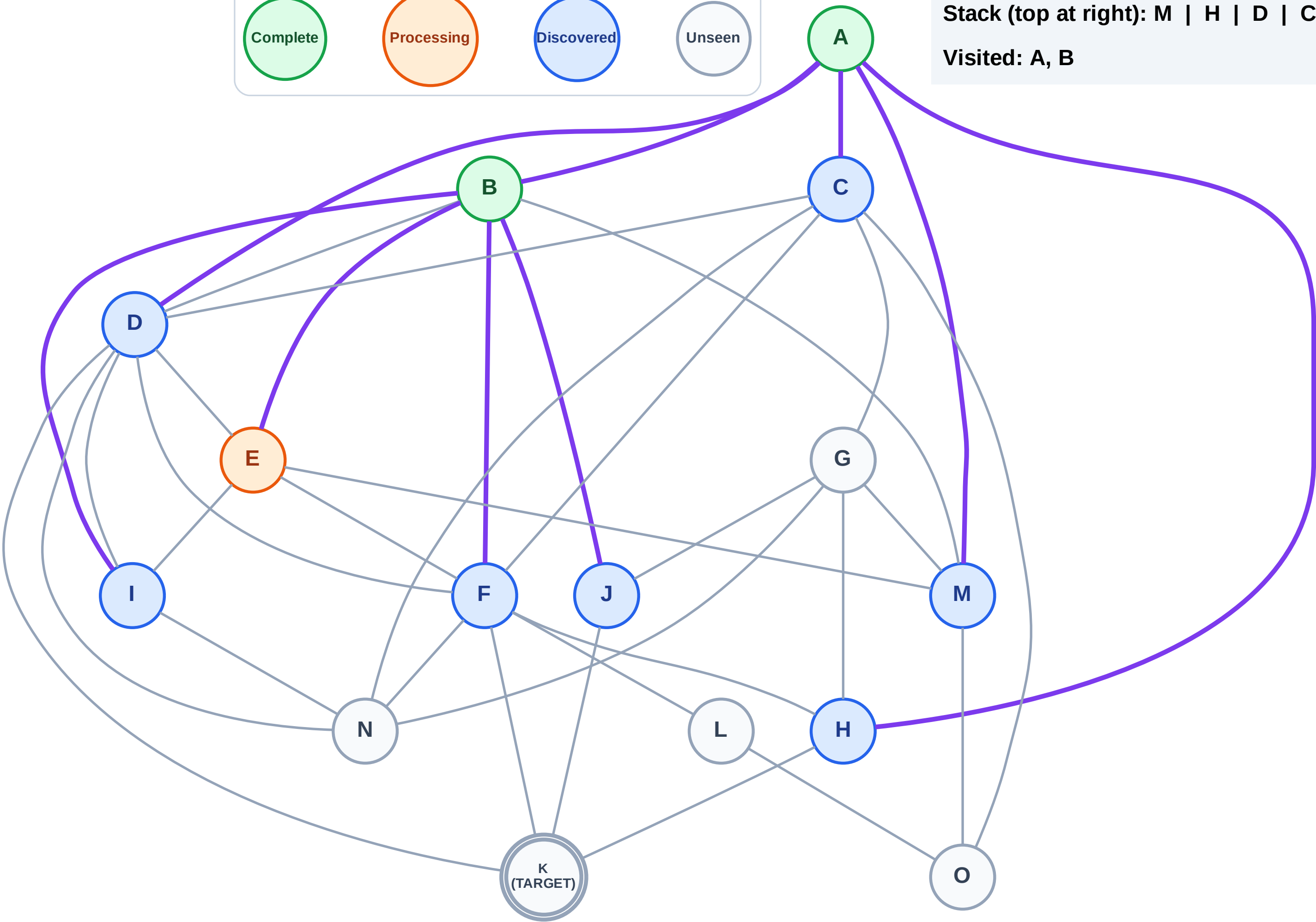
Processing

Discovered

Unseen

Stack (top at right): M | H | D | C | J | I | F

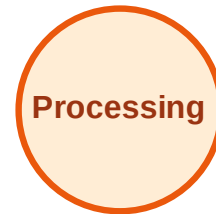
Visited: A, B



DFS Traversal

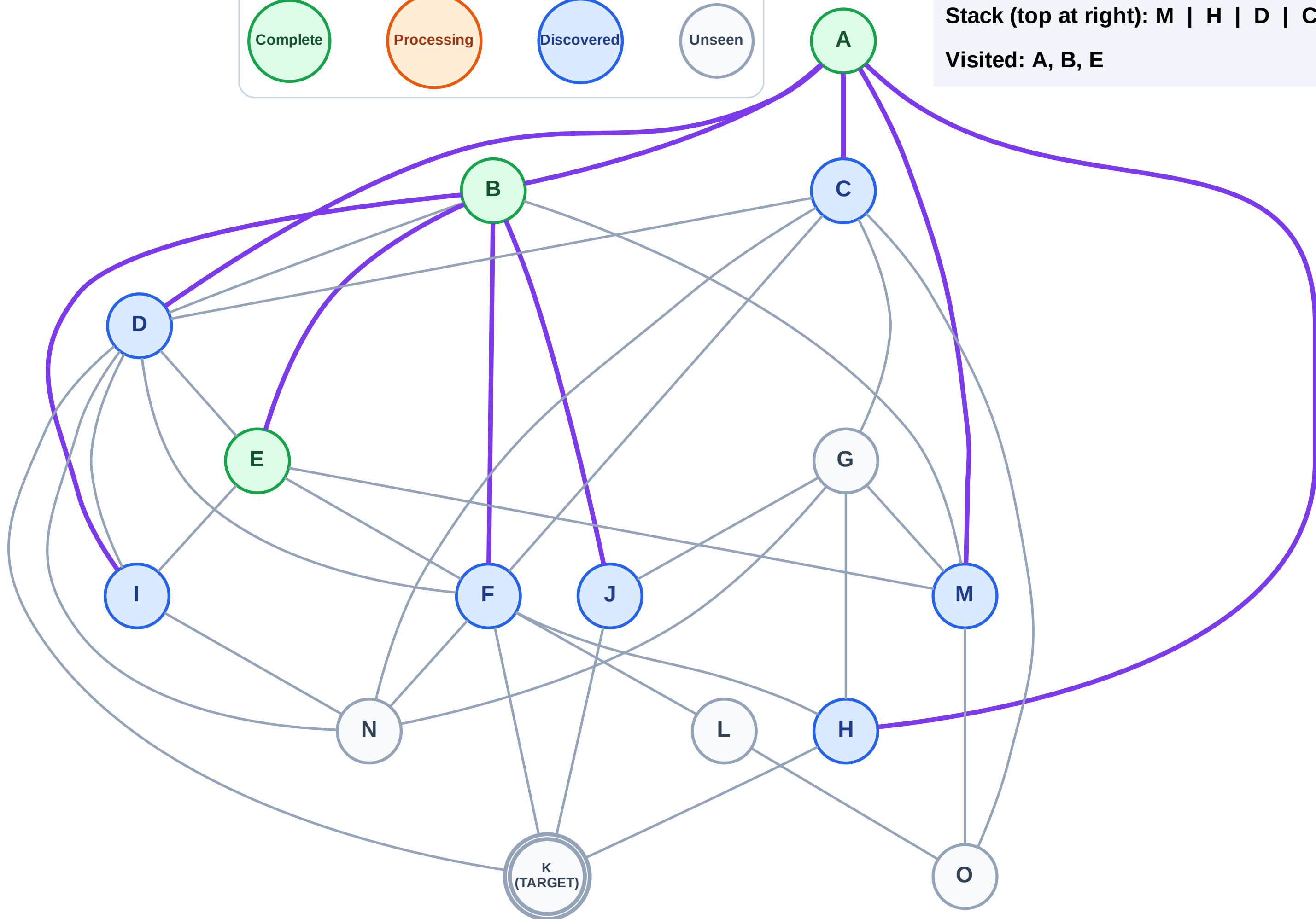
Step 7: No new neighbors from E

Legend



Stack (top at right): M | H | D | C | J | I | F

Visited: A, B, E



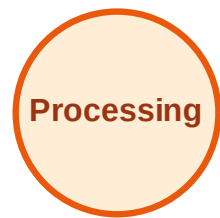
DFS Traversal

Step 8: Process node F

Legend



Complete



Processing



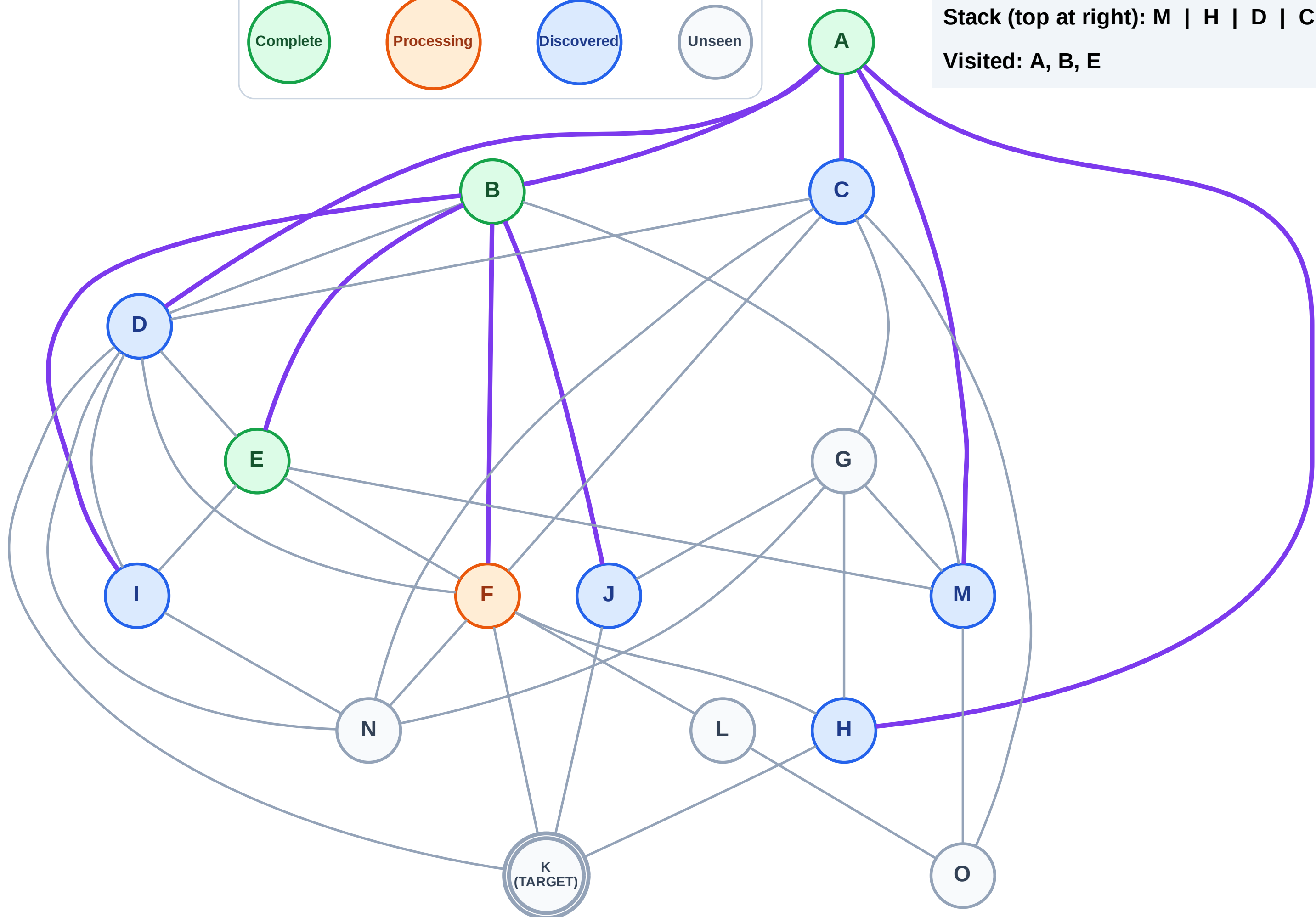
Discovered



Unseen

Stack (top at right): M | H | D | C | J | I

Visited: A, B, E



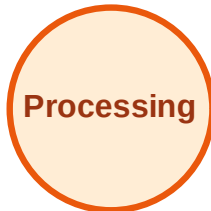
DFS Traversal

Step 9: Discover N, L, K from F

Legend



Complete



Processing



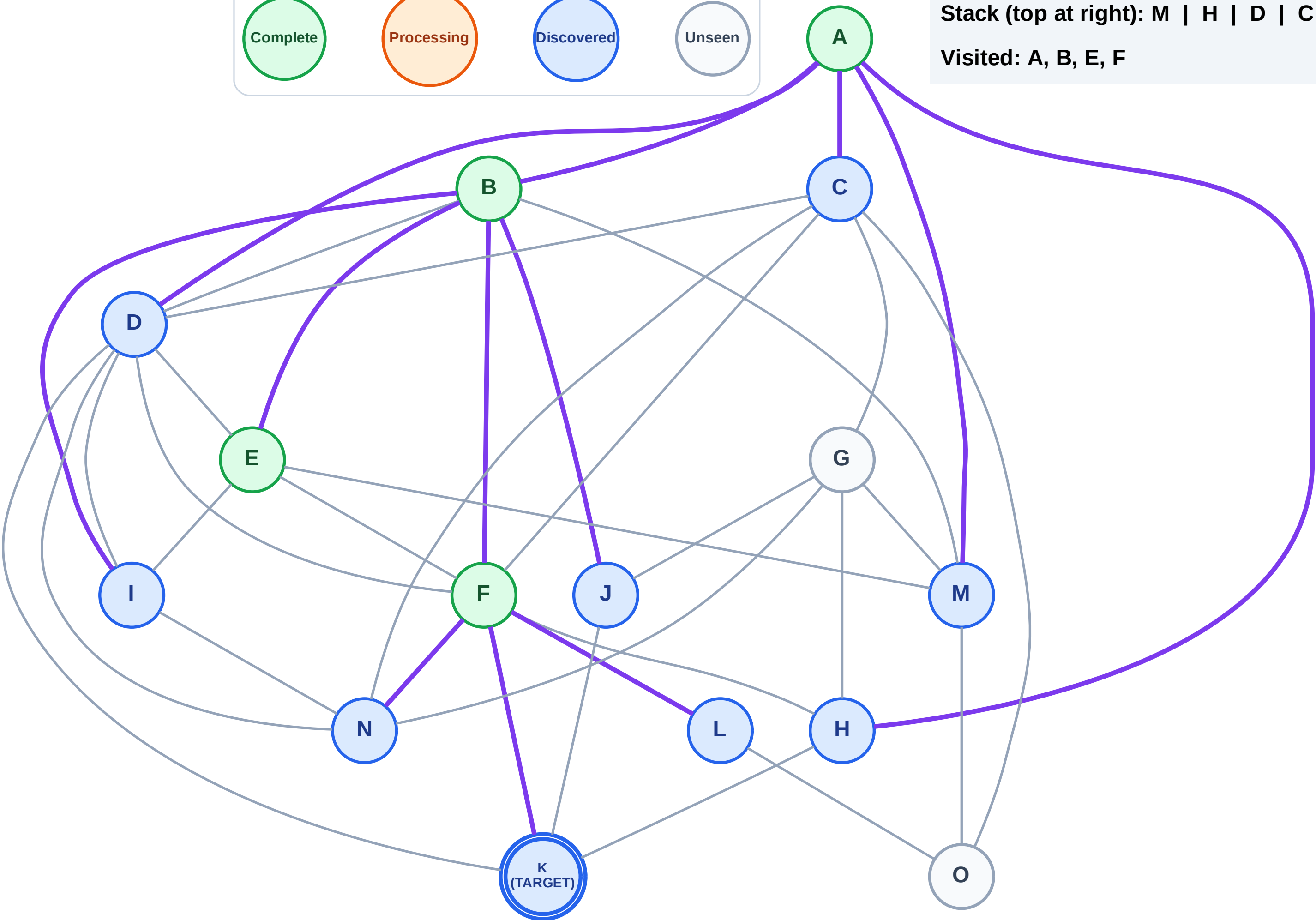
Discovered



Unseen

Stack (top at right): M | H | D | C | J | I | N | L | K

Visited: A, B, E, F



DFS Traversal

Step 10: Process node K

Legend



Complete



Processing



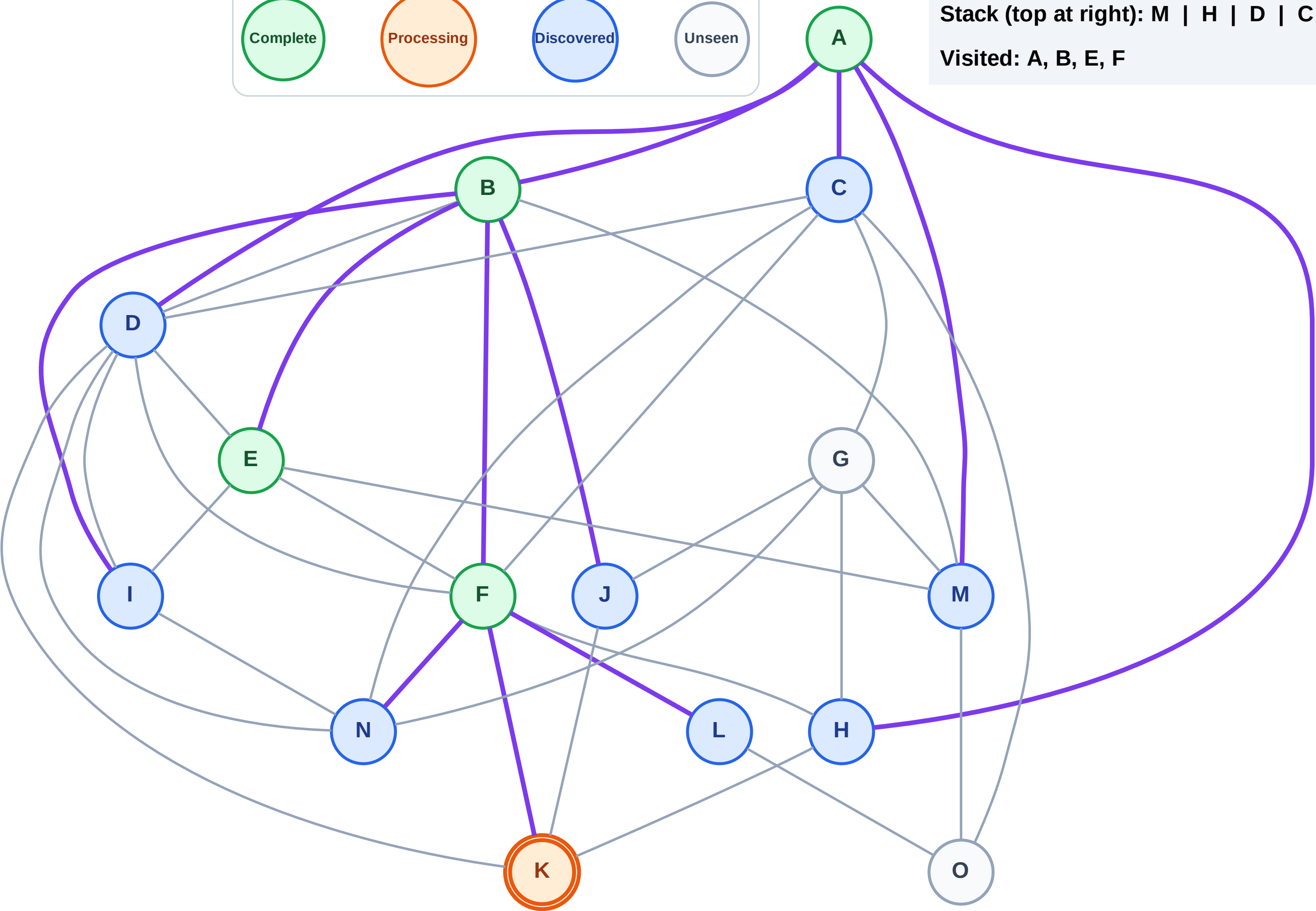
Discovered



Unseen

Stack (top at right): M | H | D | C | J | I | N | L

Visited: A, B, E, F



DFS Traversal

Step 11: TARGET NODE K REACHED! Path (3 hops): A → B → F → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): M | H | D | C | J | I | N | L

Visited: A, B, E, F, K

